Cauchy Unification: does the composition operator of a process determine the form of its scaling law?

Theory · Theory paper · Paper VII

Recursive Dynamics · The ARC Theory · Law I, the ARC Principle, a law under test

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Abstract

Why do the same scaling forms recur in systems that share no mechanism, from a mouse's heartbeat to neural networks? A rule that predicts the form of a scaling law from how a system composes, before its data are seen, would let a scaling claim be tested rather than fitted, including the power-law form that Law I of the ARC Theory, a law under test, gives recursive capability. Under a stated regularity condition, the non-zero solutions of Cauchy's four functional equations form exactly four families: linear, power law, exponential and logarithmic. Read as composition rules in a declared variable, they give a power law where depth and output both multiply and an exponential where depth adds and output multiplies. A saturating form follows from none of them: it needs added assumptions, a ceiling on output and dynamics of its own, since a ceiling alone selects no form. In a structured prediction comparison under a publicly stated rule, the composition operator of each of 25 empirical domains was classified from known physics before the author fitted candidate models and selected among them by AICc. The predicted family was the one selected in 19 of 25 domains (marginal-conditioned permutation $p = 7 \times 10^{-5}$), on a six-model candidate set with no logarithmic and no pure linear candidate, and in 14 of the 18 multiplicative and additive domains ($p = 3.0 \times 10^{-4}$). Secondary tiers are reported separately and never blended: a baseline rerun gives 15 of 20; 13 published metabolic exponents sit nearest the dimensional comparator $d/(d+1)$, seven of them still without a per-row source; 6 analytic identities check the fitter. The predictions were not timestamped in advance of their own fits and the operators were classified by the author, so the result is not proven; a replication with independent classification and a four-family test are preregistered by timestamp and await submission.

Version 3.16 states the regularity behind each solution family and the added assumptions a saturating form needs, corrects its account of the fitting method and of its March 2026 record, brings the paper's status up to date, and sets the paper in the programme's presentation standard; Appendix B lists every version and correction, with its date.

Paper VII of the ARC Theory, within Recursive Dynamics: it gives the classification behind the form of Law I, the ARC Principle, a law under test, and informs that law through the theory registration's propositions P18, that a domain's composition operator predicts its scaling family, and P9, the cross-domain form; its comparisons are dated results under the rule of March 2026, not registered verdicts, and it decides no proposition of Laws II to V (Section 12.3 gives their status). Working paper, version 3.16, 7 November 2026; first published 16 March 2026; Appendix B lists every version and correction. Recorded on OSF as *Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form*, DOI 10.17605/OSF.IO/X6WA7. About the author, founder of Recursive Dynamics, without an institution or a doctorate: michaeldariuseastwood.com/questions/who-is-michael-darius-eastwood; his book, *Infinite Architects* (first edition, 2 January 2026): osf.io/dqm3e. Licence: CC BY 4.0.

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1 Introduction

Why do scaling laws exist at all? Why does a mouse's heart beat faster than an elephant's by the same mathematical relationship that governs how cities grow, how solar panels get cheaper, and how neural networks improve with size? The standard answer is domain-specific. Biologists invoke fractal vascular networks. Physicists invoke renormalisation group flow. Economists invoke learning-by-doing. Each explanation works within its domain but offers no reason why the same mathematical forms recur across domains that share no physical mechanism. For this programme the question matters beyond any one domain. I believe advanced AI is an existential risk unless values and correction are built into how it reasons; only proof will move the world. A rule that predicts the form of a scaling law from how a system composes, before its data are seen, lets a claim about growth be tested rather than fitted, including the power-law form that Law I of the ARC Theory (the Theory of Artificial Recursive Creation) gives recursive capability; Law I, the ARC Principle, is a law under test, stated formally in Paper I.

This paper states and tests a domain-independent answer: under the conditions stated next, the form of a scaling law is determined by the composition operator of the underlying recursive process, and Cauchy's functional equations constrain which forms are possible. Stated exactly (Section 3), under a regularity condition and in a declared variable, composition in which depth and output both multiply admits only the power-law family, and composition in which depth adds while output multiplies admits only the exponential. The equations fix the family, not its exponent. A saturating form follows from none of them: the bounded case rests on added assumptions, a ceiling on output and dynamics of its own (Proposition 1), and a ceiling alone selects no form. Two domains of the comparison show what the rule asks. In radioactive decay (domain 9 of Table 2), elapsed time adds while the surviving fraction multiplies, so the composition is additive and the predicted family is the exponential; the domain's values are exact reference values, so it shows the rule at work rather than testing it (Section 11.1). The loss of neural language models against their number of parameters (domain 19; Kaplan et al. 2020) was classified as multiplicative, so the predicted family was the power law, and AICc model selection returned it.

In plain words, the result is this. In a structured prediction comparison under a publicly stated rule, the family predicted from each domain's composition operator was the one that AICc model selection returned in 19 of the 25 empirical domains, on a six-model candidate set that held no logarithmic and no pure linear candidate (Section 4.3). The operators were classified by the author, and the predictions for this cohort were not timestamped in advance of their own fits (Section 4.1); the comparison is therefore neither a proof nor a blind trial, and Sections 5 to 10 report it with its misses and with its evidence tiers kept apart. The novel contribution is not Cauchy's mathematics itself, nor the idea that invariance or one principle fixes the possible forms of laws, which earlier work states (Section 2), but the use of Cauchy's four equations together as one predictor of a domain's scaling family from its composition rule, applied before the fit and across domains, and the test of that predictor.

Our contributions are as follows.

1. A precise statement of the four classical functional equations as composition constraints: under a stated regularity condition each has one family of non-zero solutions, linear, power law, exponential or logarithmic (Theorem 1), and the family is named only once the variable in which depth composes is declared (Remark 2).
2. A proof that a saturating response satisfies none of the four equations, so a bounded composition class rests on added assumptions, a ceiling on output and a rule in a variable that the ceiling defines, which the paper names for each of its bounded candidates (Proposition 1).
3. A classification rule, applied before fitting, that predicts a scaling family from a domain's composition operator (Section 4), and a structured prediction comparison under that publicly stated rule across 50 domains in five evidence tiers, each reported separately (Sections 5 to 10).

4. A full account of what the comparison does not establish: its six misses, the two families missing from its candidate set, the timing of its predictions against their fits, and the outcomes that would defeat it (Sections 7, 12 and 13).
5. The dated record of the March 2026 extension folder and its follow-on runs, each with its kind of timestamp (Section 8.4 and Appendix C).

Section 2 sets the work against prior work. Section 3 states the four equations, their regularity, the declared variable and the bounded case, with proofs in Appendix A. Section 4 gives the prediction protocol, what was fixed before the fits, the statistical method and the limits of the candidate set. Sections 5 to 10 report the five evidence tiers, the primary result, the six misses, the published exponents, the significance framing and the analysis by group. Section 11 records the data provenance, Section 12 what this paper does and does not show, Section 13 its limitations and the outcomes that would defeat it, and Sections 14 and 15 its implications and conclusion. The appendices give the proofs, the version history, the priority record, the notation, the data sources of the 25 empirical domains and the paper's place in the programme.

2 Related work

The mathematics this paper uses is classical, and several of its ideas have specific precedents. The closest existing work falls into five groups: functional equations and the forms of laws; dimensional analysis and similarity; scale invariance; composition rules and generative processes; and the derivations of the dimensional comparator $d/(d+1)$. Each work is stated below in its own terms, and the closing paragraphs state what this paper claims as its own.

Functional equations and the forms of laws. Cauchy's functional equations, and the forms of their solutions, go back to Cauchy (1821). Aczél (1966) is the standard reference on all four, with applications to geometry, mechanics, business mathematics, probability theory and information theory; it does not treat scaling laws across domains. That invariance can fix the possible forms of a law before any data are seen is the content of Luce (1959), the closest antecedent of this paper's approach. For a continuous law relating a dependent variable to an independent one, each measured on a ratio, an interval or a logarithmic interval scale, Luce requires that an admissible transformation of the independent variable lead only to an admissible transformation of the dependent variable, and that it leave the form of the law unchanged; he lists the possible laws for each combination of scale types, among them the power function when both variables are ratio scales. Bingham and Ostaszewski (2015) state that the Cauchy functional equation is central to regular variation, and study the Goldie equation and inequality of Karamata regular variation, its counterpart in Beurling regular variation and the Gołąb-Schinzel equation; theirs is a mathematical study, with no application to scaling in physical, biological or social systems. Regular variation describes positive functions whose scale ratios $f(\lambda x)/f(x)$ converge to a power of λ as x grows, allowing a slowly varying factor at finite scale (Bingham et al. 1987). Hyers (1941) proves that the additive equation is stable under a defect bound that holds over the whole domain, the result on which Remark 3 draws.

Dimensional analysis and similarity. Buckingham (1914) shows, from the principle of dimensional homogeneity, that every complete physical equation among n quantities measured with k fundamental units is reducible to a relation among the $n-k$ independent dimensionless products of powers of those quantities; where all the other arguments of that relation are held constant, the remaining product is constant whatever the unknown relation, so any one of the quantities in it varies with the others as a product of known powers. Barenblatt (1996) sets out dimensional analysis, physical similarity theory and a general theory of scaling phenomena, and treats scaling as intermediate asymptotics: self-similar

behaviour that sets in once the influence of initial and boundary conditions has disappeared, while the system is still far from equilibrium. The book constructs self-similar solutions by dimensional analysis and treats self-similarities of the second kind separately. Both start from the dimensions of the quantities. This paper does not: it fixes the family by the rule by which a domain composes and leaves the exponent free (Section 12.2), and the d of its comparator $d/(d+1)$ is a dimension of transport geometry, not a dimension of units (Assumption 1).

Scale invariance, universality and renormalisation. That scale invariance forces a power law, by solving $p(bx) = g(b)\,p(x)$, is textbook (Newman 2005, section III.E). Newman shows that a probability distribution whose shape is unchanged when the unit of x changes, so that $p(bx) = g(b)\,p(x)$ for every b, must be a power law: setting $x = 1$ turns the condition into $p(bx) = p(b)\,p(x)/p(1)$, the multiplicative equation of Theorem 1(ii) normalised at one, which he solves for a differentiable p, and he links such scale-free behaviour to continuous phase transitions. Scaling, universality and renormalisation are linked in the review of Stanley (1999), which presents them as three of the many important themes that have served to provide the framework of much of the current understanding of critical phenomena. What this paper adds is the forward test: classifying a domain's composition before its curve is seen, and scoring the family that follows.

Composition rules and generative processes. That a composition rule, the structure of a generative process or the information a process keeps selects among exponential, lognormal and power-law forms also has precedents. Bíró (2008) proves that the infinite repetition of an arbitrary composition rule on an infinitesimal amount leads to a rule with a formal logarithm, so that the stationary distribution of energy in the thermodynamical limit is the Boltzmann-Gibbs exponential composed with that formal logarithm; interactions that depend only on the relative four-momentum squared then give kinetic energy distributions of the cut power-law (Tsallis-Pareto) form in the high-energy limit. Mitzenmacher (2004) surveys generative models for power-law and lognormal distributions: multiplicative processes, which follow Gibrat's law of proportionate effect, generate lognormal distributions, and a small change, such as a minimum that acts as a lower reflective barrier, yields a power law instead. Frank (2009) presents an informational framework for the common patterns of nature, based on the method of maximum entropy, in which those patterns follow from simple constraints on information: any aggregation of processes that preserves information only about the mean attracts to the exponential pattern, and any that preserves information only about the geometric mean attracts to the power-law pattern; his examples include the age of disease onset, rates of amino acid substitution and the composition of ecological communities. The idea that one principle selects the common form of patterns across unrelated fields is therefore stated in Frank's work, and the idea that invariance under changes of scale restricts the possible forms of a law is stated in Luce's. The works of Newman, Bíró, Mitzenmacher and Frank concern probability distributions. Luce's concerns laws between measured variables, with the form fixed by the scale types of the variables; this paper fixes the family of a law by the rule by which its domain composes, classified from the domain's known physics before the fit.

Derivations of the dimensional comparator. The expression $d/(d+1)$, or a special case of it, is reached in earlier work through different frameworks. West et al. (1997) derive allometric scaling relations, including the three-quarter power law for metabolic rates, from a model of how essential materials are transported through space-filling fractal networks of branching tubes, assuming that the energy dissipated is minimised and that the terminal tubes do not vary with body size; beyond mammalian circulation, their model predicts structural and functional properties of vertebrate cardiovascular and respiratory systems, plant vascular systems, insect tracheal tubes and other distribution networks. The empirical exponent is

debated (the note on the three-quarter exponent in Section 8.1). Banavar et al. (1999) derive a general relationship between size and flow rates in networks with local connectivity, which accounts in a general way for the quarter-power allometric scaling of living organisms, with metabolic rate scaling as mass to the power $D/(D+1)$ in the most efficient networks filling D dimensions; their theory also predicts scaling relations applicable to all efficient transportation networks, which they verify with observational data on river drainage basins. In two models of resource distribution networks, Banavar et al. (2010) show that the exponent of metabolic rate against body mass is $2/3$ if the velocity of flow stays constant and reaches its maximum, $3/4$, if velocity scales with its maximum exponent, $1/12$, and that quarter-power scaling can arise with no underlying fractality; the geometry of the network alone therefore does not select one exponent. Demetrius (2003, 2006) and Demetrius and Tuszynski (2010) reach allometric scaling through quantum statistics: the 2010 model, of coupled energy-transducing oscillator networks embedded in a d-dimensional space, joins the chemiosmotic theory of energy transduction to the Debye model of the thermal properties of solids, and gives the exponent $d/(d+1)$ in its quantum regime ($1/2$, $2/3$ and $3/4$ for $d = 1$, 2 and 3) and isometry in its classical regime, on physical assumptions different from those of vascular branching. In Bettencourt (2013), urban land area scales with population to the power $D/(D+H)$, where D is the dimension of the space and H the fractal dimension of travel paths; with $D = 2$ and $H = 1$ this is $2/3$, and the expression takes the form $D/(D+1)$ only when $H = 1$. The same framework gives socioeconomic outputs that grow superlinearly with population, with exponent $7/6$ in that case. Zhao (2022) derives an $n/(n+1)$ power scaling law for an n-dimensional system from producer-consumer dynamics, and places biomass production, eukaryote growth, mammalian brain size and the distribution of public facilities in cities in a new universality class whose dimensionality determines its growth scaling. Earlier versions of this paper also listed a derivation of metabolic scaling exponents by He and Chen (2003) from optimal vascular network geometry with flow conservation constraints; the one He and Chen record of 2003 at Crossref (He and Chen 2003) is *Effects of size and pH on metabolic rate*, in the *International Journal of Nonlinear Sciences and Numerical Simulation*; it carries no abstract, and on 3 October 2026 it could not be confirmed that this record is that work, so nothing here rests on it. He and Zhang (2004) state, of brain cells, that the fractal geometry of the cell "leads to a 4/5 allometric scaling law for human brain"; they give a physical explanation for the fourth dimension of life for three-dimensional organisms and the fifth dimension for the human brain. Maino et al. (2014), within dynamic energy budget theory, "derive an equation for the scaling of metabolic rate for adult heterotrophs without relying on optimisation arguments or implying cellular nutrient supply limitation"; their abstract states no exponent. Geoffrey West's book *Scale* (West 2017) discusses scaling in organisms, cities and companies, and does not mention Cauchy's functional equations.

What is and is not claimed here. These derivations reach the expression $d/(d+1)$, in its general form or as a special case, through different frameworks: fractal branching networks, the geometry of efficient transportation networks, quantum metabolism, urban scaling (as $D/(D+H)$ with $H = 1$), producer-consumer dynamics, optimal vascular geometry and dynamic energy budget theory; of these, the He and Chen item could not be confirmed at a registry on 3 October 2026, and the abstract of Maino et al. (2014) states no exponent. They are not counted here as separate confirmations of the comparator. This paper's hypothesis is that where such a derivation reaches $d/(d+1)$, it rests on three conditions: multiplicative composition, which Theorem 1 constrains to the power-law family; d-dimensional space-filling transport geometry; and a conservation or optimisation constraint on resource flow (Assumption 1). On this hypothesis, Cauchy's equation accounts for the family and the shared transport conditions for the value. A common numerical fraction does not by itself identify a common mechanism, as the framework paper now states (*On the Origin of Scaling Laws*, version 3.14), so the hypothesis is tested derivation by derivation, by whether each one's premises include the three conditions.

The mathematical theorems are 200 years old. The biological data are 90 years old. What this paper claims as its own is neither the formula $d/(d+1)$, which is well established, nor Cauchy's mathematics, but the hypothesis just stated and the predictor of the next paragraph, which Sections 4 to 10 put to a structured test.

Because the work conceded above already uses invariance, or one principle, to fix the possible forms of laws and patterns across fields, the novelty this paper claims is narrower. The search recorded in this paper's public prior-art record (first committed on 27 June 2026 and last revised on 21 October 2026) found no publication that uses Cauchy's four equations together as one predictor of a domain's scaling family from its composition rule, applied before the fit and across domains, and tests that predictor as this paper does; nor did it find an earlier source that names Cauchy's multiplicative equation in accounting for the common form of the derivations above. Readers are invited to submit prior work; a genuine antecedent that does this defeats the priority claim of this paper.

3 The theory

Cauchy's four functional equations, additive, multiplicative, exponential and logarithmic, each have one family of non-zero solutions under a stated regularity condition (Theorem 1). We argue that this 200-year-old mathematical result has a physically testable consequence: under that regularity condition, it forces a power law where composition is multiplicative and an exponential where depth adds while output multiplies. Where output is bounded it forces no form. A saturating response satisfies none of the four equations (Proposition 1): a saturation curve compatible with a bound rests on added assumptions, a ceiling on output and dynamics of its own, and the bound alone selects no form (Section 3.3). Once the variable in which depth composes is declared (Remark 2), the family is determined by the composition operator of the underlying recursive amplification process.

3.1 Cauchy's four equations

Augustin-Louis Cauchy established in the early nineteenth century that the continuous solutions of four functional equations form four one-parameter families (Cauchy 1821). The solutions are families, not unique functions, and the same families follow under conditions weaker than continuity (Aczél 1966, chapter 2). Theorem 1 states the four equations with their domains and the regularity each needs.

Theorem 1 (Cauchy's four equations under regularity)**.** *Let f satisfy one of the following equations for all arguments in its domain, and suppose f is continuous, or Lebesgue measurable, or bounded on some interval of positive length.*

(i) $f : \mathbb{R} \to \mathbb{R}$, $f(x+y) = f(x) + f(y)$*: then* $f(x) = cx$.

(ii) $f : (0,\infty) \to (0,\infty)$, $f(xy) = f(x)f(y)$*: then* $f(x) = x^c$.

(iii) $f : \mathbb{R} \to (0,\infty)$, $f(x+y) = f(x)f(y)$*: then* $f(x) = a^x$*, with* $a = e^c > 0$.

(iv) $f : (0,\infty) \to \mathbb{R}$, $f(xy) = f(x) + f(y)$*: then* $f(x) = c \log x$.

Here c is a real constant, the logarithm is natural, and each stated function satisfies its equation. If the targets in (ii) and (iii) are taken to be all real numbers, the only further solution is $f \equiv 0$. Restricting the arguments of (i) and (iii) to $(0,\infty)$ leaves the same families.

Proof in Appendix A.1.

These are not four independent results. They are four faces of one constraint: the requirement that a function be compatible with a binary operation. Under regularity (continuity, or Lebesgue measurability, or boundedness on an interval of positive length), a function that carries one of the two structures, addition

on the real numbers or multiplication on the positive real numbers, into one of the two falls into exactly four families: linear ($f(x) = cx$), power law ($f(x) = x^c$), exponential ($f(x) = a^x$) and logarithmic ($f(x) = c \log x$). This exhausts the homomorphisms. It does not exhaust scaling laws, because most systems are not homomorphisms of either structure: saturating behaviour, curvature on log axes and multi-regime crossovers all lie outside the grid, and the grid says nothing about them. A critical power law can have the power-law form; Theorem 1 neither excludes it nor derives its critical exponent or physical mechanism. The additive equation yields the linear form $f(x) = cx$; the multiplicative equation yields power laws; the exponential equation governs exponential growth and decay; and the logarithmic equation governs the map $c \log x$, which is multiplicative in input and additive in output; its increments diminish, but it is unbounded, so it is not a saturating family.

Remark 1. Without a regularity condition the additive equation has other solutions, which are not Lebesgue measurable and are unbounded on every interval (Aczél 1966); through the substitutions of Appendix A each of the other three equations has corresponding solutions. A rule observed only at some arguments, for example only under doubling, does not establish the equation (Eastwood 2026f).

Remark 2. The family label depends on the declared variable. With $s = \log R$, the multiplicative equation in R becomes the exponential equation in s, and $R^c = e^{cs}$; a power law in R is an exponential in $\log R$. Normalisation matters as well: $Y(x) = Ax^c$ satisfies $Y(xy) = Y(x)Y(y)/A$, so the equation applies to $Y/Y(1)$ (Eastwood 2026g). The classification therefore fixes the variable in which depth composes, and its baseline, before it names a family.

Appendix A.3 works the change of variable.

3.2 The prediction

Consider any system where a quantity U grows with recursive depth R, measured in a declared variable (Remark 2). The system's *composition operator* determines how consecutive applications combine, and Definition 1 states the two exact rules.

Definition 1. A response $U(R) > 0$ to recursive depth R, in the declared variable R, has *multiplicative composition* if $U(R_1R_2) = U(R_1)U(R_2)/U(1)$ for all $R_1, R_2 > 0$, and *additive composition* if $U(R_1{+}R_2) = U(R_1)U(R_2)/U(0)$ for all $R_1, R_2 \geq 0$. The first rule needs U for $R > 0$; the second needs it for $R \geq 0$, with $R = 0$ the baseline from which depth is counted.

- **Multiplicative composition** (depth multiplies, output multiplies): the normalised response $U/U(1)$ satisfies Cauchy's multiplicative equation, so under the regularity of Theorem 1 it forces $U \propto R^c$, a power law.
- **Additive composition** (depth adds, output multiplies): the normalised response $U/U(0)$ satisfies Cauchy's exponential equation, so under the same regularity it forces $U \propto a^R$, an exponential.

Remark 3. Where composition holds only approximately, as the sign $\sim$ in Section 2.2 of version 3.15 wrote it, a family follows only under a stated error bound that holds over the whole domain: for example, if a continuous $F : (0, \infty) \to (0, \infty)$ has a logarithmic defect $|\log F(xy) - \log F(x) - \log F(y)|$ bounded by ε for every pair of arguments, then $e^{-\varepsilon}x^c \leq F(x) \leq e^{\varepsilon}x^c$ for a unique real c (Hyers 1941; Eastwood 2026g). Finitely many noisy observations cannot establish such a bound, so the comparison in Sections 6 to 10 compares predicted and selected families; it does not establish the equation in any domain.

3.3 Bounded composition

Bounded composition (output constrained to $[0, K]$): a bound selects no form. A bound and convergence are different premises: a bounded response need not converge at all, and one that rises monotonically

with depth converges to a limit no higher than K, at a rate and along a path that the bound does not fix (Eastwood 2026g, Section 5). The saturation curves of the candidate set of Section 4, the logistic, Hill, Michaelis-Menten and saturation-exponential forms, are such responses; they form a candidate set, not an exhaustive class. None of them satisfies any of the four equations in R: each rests on the ceiling K, an assumption added to the composition rule, and satisfies one of the equations only in a variable that the ceiling defines (Proposition 1). Which of them, if any, a bounded domain follows is fixed by dynamics of its own, not by the bound.

Proposition 1 (Saturating responses lie outside the four equations)**.** *Let f be a non-constant solution of one of the four equations of Theorem 1, with the regularity it assumes. Then, as $x \to \infty$, $f(x)$ tends to $+\infty$, to $-\infty$ or to 0. Hence a non-constant response $U(R)$ that tends to a finite ceiling $K \neq 0$ as $R \to \infty$ satisfies none of the four equations in R. Each bounded candidate of Section 4 satisfies one of them only in a variable that the ceiling defines: the saturation exponential $U = K(1 - e^{-cR})$ makes the deficit $K - U = Ke^{-cR}$ an exponential in R; the logistic $U = K/[1 + e^{-\lambda(R-R_0)}]$ makes the log-odds $\log[U/(K-U)] = \lambda(R - R_0)$ linear in R measured from R_0; and the Hill and Michaelis-Menten forms $U = KR^n/(k^n + R^n)$ ($n = 1$ for Michaelis-Menten) make it $n \log(R/k)$, logarithmic in R/k. The ceiling K and the choice of transformed variable are assumptions added to the composition rule; they do not follow from any of the four equations.*

Proof in Appendix A.2.

The framework paper reaches the same conclusion: "Saturation therefore requires its own dynamics and is not a fifth Cauchy cell" (Eastwood 2026g, Section 5).

The prediction. If the composition operator of a system is known, in a declared variable, and its response obeys the corresponding rule of Definition 1 for every pair of depths, the family of its scaling law follows before any data are seen: a power law where composition is multiplicative and an exponential where it is additive (Theorem 1). Where output is bounded, the prediction is a saturation curve, which rests on an assumed ceiling and on dynamics that the composition rule does not supply (Proposition 1). Whether a domain's response obeys such a rule is the premise that the classification of its composition operator supplies, and the comparison of Sections 6 to 10 puts that premise at risk against rival families on each domain's data. The family is predicted with no curve fitting; its parameters, the exponent, the base and the ceiling, are not, and they are fitted. Where composition holds only approximately, the prediction carries the condition of Remark 3.

4 Prediction protocol

We present a structured prediction comparison across 50 domains in five evidence tiers (Section 5), among them six analytic identities that check the fitter. Each empirical domain followed the same sequence: known physics fixed the composition operator before any fitting began, the operator class then set which scaling family to expect, published or standard reference data were loaded, and that expectation was finally put to the test by model fitting with AICc-based model selection, run by the author after the classification rule was public (16 March 2026, 23:33 UTC).

For each domain:

1. **Classify the composition operator from known physics** (before fitting). Determine whether the system's recursive amplification process combines multiplicatively, additively, or with a bound.
2. **Predict the scaling family** from the operator class: by Cauchy's equations, a power law for a multiplicative operator and an exponential for an additive one (Theorem 1); for a bounded operator,

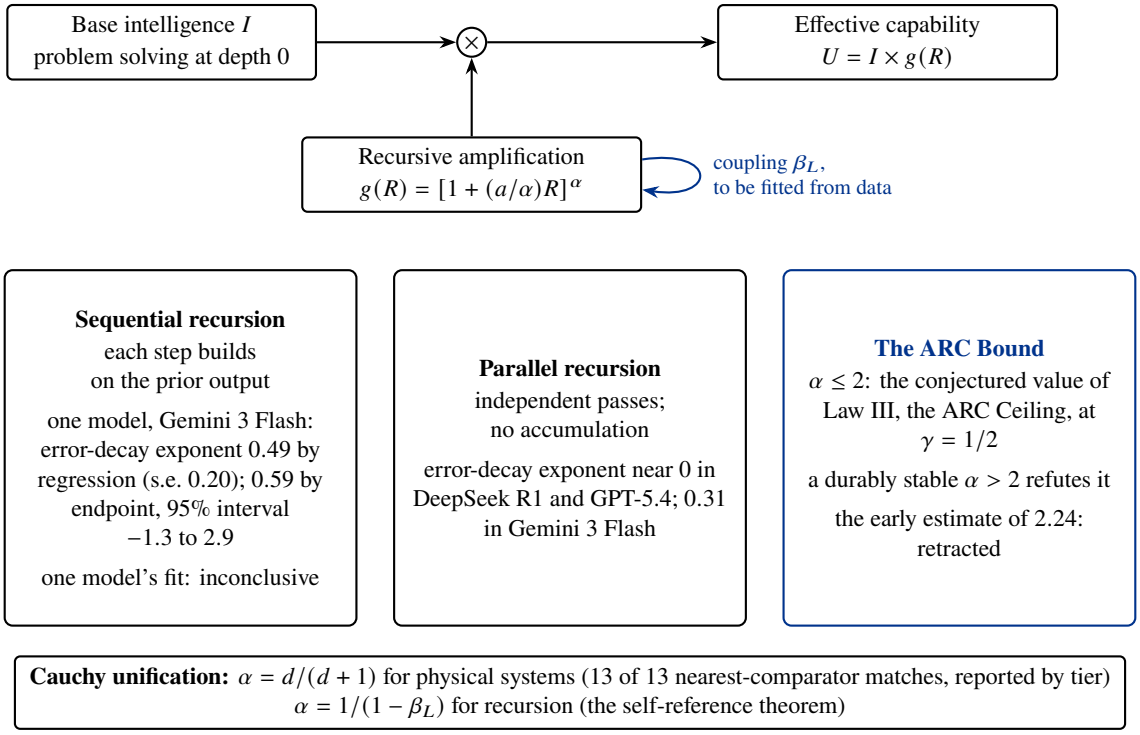


Figure 1. The ARC architecture and the Cauchy unification, redrawn in this version. *Top:* base intelligence I, problem solving at depth 0, is multiplied by the recursive amplification $g(R) = [1 + (a/\alpha)R]^{\alpha}$, with $g(0) = 1$, rate constant a, and an exponent α set by the self-referential coupling β_L, which is to be fitted from data; the product is the effective capability $U = I \times g(R)$, the architecture of Law I, the ARC Principle, a law under test (Eastwood 2026h). *Cards:* the two left cards give Paper II's error-decay exponents, which describe how error falls as recursive depth grows (Eastwood 2026i). For sequential recursion, on the one model whose data scaled smoothly, Gemini 3 Flash, the exponent is 0.49 by regression (standard error 0.20) and 0.59 by endpoint estimate, whose bootstrap 95% interval runs from −1.3 to 2.9: one model's fit, and inconclusive. For parallel recursion it is near zero in DeepSeek R1 and GPT-5.4 and 0.31 in Gemini 3 Flash; Paper II gives the other models' rows. An error-decay exponent is a different observable from the capability exponent α of $g(R)$, and reading one as the other needs a stated metric bridge, which this figure does not supply. The right card is the ARC Bound, $\alpha \leq 2$, the conjectured value of Law III, the ARC Ceiling, $\alpha_{\text{crit}} = 1/(1-\gamma)$, at a correction-leverage exponent $\gamma = 1/2$. Law III is a law under test; a durably stable $\alpha > 2$ refutes the bound, and the early estimate of 2.24 is retracted (Paper IX, Section 7; Eastwood 2026k). *Bottom:* under regularity, the four functional equations of Cauchy (1821) have four solution families, linear, power law, exponential and logarithmic (Theorem 1). For physical systems the dimensional comparator is $\alpha = d/(d+1)$, under the conditions of Assumption 1: 13 of 13 nearest-comparator matches at the taxon grain, reported by tier and never blended, and the largest modern mammalian dataset gives 0.687, with a 95% confidence interval from 0.674 to 0.701, which excludes 0.750 (Section 8.1). For recursion, $\alpha = 1/(1-\beta_L)$ is the self-reference theorem of the Foundational paper (Eastwood 2026d), where it is the asymptotic exponent of that paper's growth model, with a fixed rate constant $a > 0$ and $0 \leq \beta_L < 1$ (its theorem on the scaling exponent). It is one of the two routes to the value two; the second is the ARC Ceiling, and whether the two agree away from $\beta_L = \gamma = 0.5$ is a drafted test, preregistered by timestamp and awaiting human submission. In 19 of the 25 empirical domains of Section 6 the predicted family was the one selected (permutation $p = 7 \times 10^{-5}$), on the six-model candidate set, which held no logarithmic and no pure linear candidate (Section 4.3). Appendix B lists the changes made in redrawing the figure, and Appendix B.4 records the earlier artwork's labels.

a saturation curve, which rests on an assumed ceiling and on dynamics of its own rather than on Cauchy's equations (Proposition 1).

3. **Load published or standard reference data.**
4. **Fit candidate models** (power law, exponential, logistic, Hill, Michaelis-Menten, saturation-exponential) to the data: the four saturation curves by nonlinear least squares on the response, and the power law and the exponential by linear least squares on the logarithm of the response. For the muscle force-velocity domain (domain 15), whose predicted model was a hyperbola, a seventh candidate, the hyperbolic decay $a/(b + R)$, counted in the bounded family, was also fitted.
5. **Select the best model by AICc** (corrected Akaike information criterion) and compare the prediction against the selected model's family.

4.1 What was fixed before the fits

The operator classification is made before the fitting procedure runs. However, the predictions, data, and fitting logic all reside in a single author-written script, and no registry submission exists for this test. The predictions for this cohort are not timestamped in advance of their own fits: the classification rule reached the public repository at 23:33 UTC on 16 March 2026 and the fifty per-domain assignments were committed together with their results at 23:54 that night, attestation class public repository commit date for both, so predicted families before fits is true of the procedure and not of the timestamp. The advance per-domain timestamp in this record belongs to the twelve-domain extension, whose manifest was committed at 00:19:19 UTC on 17 March 2026, twenty-three minutes before its fits, and to the book's two prediction appendices in print on 2 January 2026, before the papers and the programme existed (Eastwood 2026a). What was not fixed in advance here was the fitting protocol, and that gap is closed prospectively by the staged tests, the fourth-cell test and the 80-domain expansion test, both drafted and awaiting human submission. This test should therefore be understood as a structured prediction comparison, not a blind trial. A replication with independent operator classification has since been drafted, as an externally classified confirmatory extension of this paper, and it awaits human submission. The replication and the fourth-cell test are preregistered by timestamp, each first anchored on 12 August 2026 at Bitcoin block 962066.

A server-dated deposit of the twelve-domain extension's files exists and is not claimed as covering this result. On 17 March 2026, between 02:06:51 and 02:07:05 UTC, a prediction protocol, an extension manifest and a checksum file were deposited to the public Open Science Framework component x6wa7. Each carries an OSF-assigned creation timestamp, and each has a modification timestamp identical to its creation timestamp. That is a dated third-party deposit rather than an OSF Registration, which is a distinct frozen and schema-bound object this component does not carry; the API reports registration: false and this paper says so rather than blurring it. The manifest is the twelve-row manifest of the extension of Section 8.4, deposited after that extension's run (00:42:41 UTC) and after the author relabelled the run (00:56 UTC), so the deposit corroborates the domain list and the date on which it existed, not the prospective standing of the run (Eastwood 2026m); the extension's advance timestamp is its public commit of 00:19:19 UTC (Appendix C). Independent replication is needed.

4.2 Statistical method

The v2 test uses proper AICc (corrected Akaike information criterion) computed from residual sum of squares, sample size, and parameter count, not the R^2-based approximation used in the legacy v1 test. The v1 approximation,

$$\mathrm{AIC} \approx n \ln(1 - R^2) + 2k, \tag{1}$$

was adequate as a heuristic but should not have been described as strict AIC in the standard statistical sense. The v2 rewrite corrects this.

Every candidate's AICc is computed from the residual sum of squares of the response itself. The power law and the exponential are scored at the parameters of their fit on the logarithmic scale, which do not minimise that sum, so their AICc is never lower than at their own least-squares optimum on the response: the procedure can only disfavour them against the saturation curves. That is conservative for the matches among the 18 multiplicative and additive domains, may have contributed to the four multiplicative misses, and can favour the bounded family in the seven bounded domains; whether it changed any selection has not been checked.

The primary significance test is a marginal-conditioned permutation test: holding both the predicted and the observed family marginals fixed and permuting the pairing 400,000 times, the probability of 19 or more matches is $p = 7 \times 10^{-5}$. The expected match count under this null is 9.5 of 25, not the 8.33 a uniform null gives, because power laws are simply common among the observed fits. A one-sided binomial test against a uniform three-family null gives $p = 1.56 \times 10^{-5}$ and is retained as a sensitivity check; it is the weaker null and is not used as the headline. Sigma equivalences are not quoted because they shift with the choice of null. The script that computes this test, which parses the published Table 2 directly so that it cannot drift from this paper, is held in a private repository and is not yet public; the test can be recomputed from that table's Predicted and AICc best columns, which fix both marginals.

4.3 A model-set limitation this paper must state against itself

Section 3.1 names four Cauchy families. The candidate model set of this section contains six models (seven for domain 15, step 4): a power law and an exponential, which span two of those families, and four saturation curves, which lie outside the four equations (Proposition 1). Two of the four continuous Cauchy solutions have no candidate of their own: the logarithmic form $c \log x$ and the pure linear form cx. The power-law candidate contains the linear form as its case of exponent one, and cx satisfies the multiplicative rule of Definition 1, so a linear response is fitted as a power law and counts as a match wherever its domain was classified as multiplicative. One consequence of the missing logarithmic candidate runs each way. A genuinely logarithmic domain could not have been classified correctly, because the fitter had no logarithmic candidate to select, so some of the six misses may have been guaranteed misses; and some of the nineteen hits may be logarithmic relations that a power law fits acceptably over a finite range. Every hit count in this paper therefore carries this qualifier, and none may be quoted without it. A confirmatory study that re-runs the comparison with the full four-family candidate set, with per-cell outcomes specified in advance, has been drafted and preregistered by timestamp (Section 4.1), and awaits human submission as a draft registration (the four-family logarithmic-cell test in the programme's Dated Prediction Register (Eastwood 2026l), alongside the fresh-domain extension study). Until that re-run reports, the numbers below are the six-model result, exactly as computed, and nothing stronger.

The 19 of 25 is the record of what the classification rule of March 2026 returned on this candidate set. No count under the corrected classification is printed here: such a count comes from a rerun of 11 August 2026, and the author's registration update of 13 September 2026 records that an analysis run that day was not requested by him and that he has not examined its outputs (Eastwood 2026m).

5 Evidence tiers

The 50-domain suite comprises five evidence tiers of decreasing strength (Table 1). These tiers are never combined into a single blended number.

Table 1. The five evidence tiers, in decreasing strength, each reported separately and never blended. The primary count is on the six-model candidate set of Section 4.3. Source: the counts and the binomial probabilities are those of the suite's results file in the public repository, results_50_domain_validation.json, and the permutation probability is described in Section 4.2. The baseline-20 rerun re-fits domains 1 to 20 of the primary tier and adds no domain, so the suite's 50 domains are the 25 of the primary tier, the 13 and 6 published exponents and the 6 analytic identities. Sections 6 to 8 report each tier.

Tier	Description	Count	Result	Significance
Primary	Empirical curve-fit (AICc)	25	19 of 25 (six-model set)	$p = 7 \times 10^{-5}$ perm.; 1.56×10^{-5} binomial
Secondary	Baseline-20 rerun (AICc)	20	15 of 20	$p = 1.67 \times 10^{-4}$
Secondary	Published exponents (direct)	13	13 of 13	Nearest comparator match
Noted	Published exponents (provisional)	6	3 of 6	Contested organisms
Noted	Analytic identities	6	6 of 6	Tautological confirmation

6 Primary result: 25 empirical curve-fit domains

Table 2. The 25 empirical curve-fit domains: the composition operator assigned to each, the family that operator predicts, the model with the lowest corrected Akaike information criterion (AICc best), and whether that model lies in the predicted family. Each domain's data source is listed in Appendix E.

#	Domain	Operator	Predicted	AICc best	Match
1	Kleiber's Law (Metabolic Scaling)	Multiplicative	Power law	Saturation-exp	No
2	Urban Scaling (GDP vs Population)	Multiplicative	Power law	Power law	Yes
3	Species-Area (Galapagos)	Multiplicative	Power law	Hill	No
4	Wright's Law (Solar PV)	Multiplicative	Power law	Power law	Yes
5	Heaps' Law (Vocabulary)	Multiplicative	Power law	Power law	Yes
6	Zipf's Law (Word Frequency)	Multiplicative	Power law	Michaelis-Menten	No
7	Learning Curve (Cigar Rolling)	Multiplicative	Power law	Power law	Yes
8	Moore's Law (Transistor Count)	Additive	Exponential	Exponential	Yes
9	Radioactive Decay (P-32)	Additive	Exponential	Exponential	Yes
10	Gutenberg-Richter (Earthquakes)	Additive	Exponential	Exponential	Yes
11	Bacterial Growth (*E. coli*)	Bounded	Bounded	Logistic	Yes
12	O_2-Haemoglobin Curve	Bounded	Bounded	Hill	Yes
13	Epidemic SIR (Ebola 2014)	Bounded	Bounded	Hill	Yes
14	Amdahl's Law (CPU Scaling)	Bounded	Bounded	Michaelis-Menten	Yes
15	Muscle Force-Velocity (Hill 1938)	Bounded	Bounded	Exponential	No
16	Facebook MAU Growth	Bounded	Bounded	Logistic	Yes
17	Brownian Diffusion (MSD)	Multiplicative	Power law	Power law	Yes
18	Horton's Law (Stream Numbers)	Additive	Exponential	Exponential	Yes
19	Neural Scaling Laws (LLM Loss)	Multiplicative	Power law	Power law	Yes
20	Time Crystal Order (Rydberg Gas)	Bounded	Bounded	Power law	No
21	Stellar Mass-Luminosity	Multiplicative	Power law	Logistic	No
22	Heart Rate vs Body Mass	Multiplicative	Power law	Power law	Yes
23	Rent's Rule (VLSI Pin Count)	Multiplicative	Power law	Power law	Yes
24	Taylor's Power Law	Multiplicative	Power law	Power law	Yes
25	Hack's Law (Stream Length)	Multiplicative	Power law	Power law	Yes

Primary result, on the six-model set. Under strict AICc-based model selection, on the six-model candidate set, which contained no logarithmic and no pure linear candidate (Section 4.3), the AICc-best family matched the Cauchy-predicted family in 19 of the 25 empirical curve-fit domains of Table 2. No count from the rerun of 11 August 2026 is printed: the author did not request that analysis and has not examined its outputs (Section 4.3). The primary significance statistic is a permutation test conditioned on both the predicted and the observed family marginals: $p = 7 \times 10^{-5}$. A one-sided binomial test against uniform $1/3$ chance, that is, uniform random assignment to the three families present in this candidate set, gives $p = 1.56 \times 10^{-5}$, roughly 1 in 64,000, and is retained as a sensitivity check; under the four-family grid the same result gives $p = 1.27 \times 10^{-7}$, roughly 1 in 7.9 million, so the figure reported here is the conservative one. On the strictly Cauchy-derived subset alone (the 18 multiplicative and additive domains, excluding the bounded family, which rests on added assumptions; Section 3.3) the count is 14 of 18, at permutation $p = 3.0 \times 10^{-4}$: the headline does not depend on the bounded family. Two of the 19 matches, radioactive decay (domain 9) and Amdahl's law (domain 14), use data computed from the formula itself, and Amdahl's formula is of the Michaelis-Menten form that was selected (Section 11.1): like the analytic identities of Section 8.3, they check the fitter rather than test the rule, and both are counted in the 19, the first also in the 14.

This is strong enough to publish as a structured prediction comparison under a publicly stated rule rather than a per-domain advance timestamp. It is not strong enough to claim as proven. Independent replication with preregistered operator classification is the necessary next step.

6.1 Baseline-20 rerun

As context, the original 20 baseline domains were rerun with the new proper AICc fitter. The result is 15 of 20 ($p = 1.67 \times 10^{-4}$). This is not directly comparable to the v1 result of 14 of 20, because the fitter changed (R^2-based AIC approximation to proper AICc) and some model selections shifted accordingly. The baseline-20 rerun is reported for transparency, not as evidence that the new fitter is more or less conservative than the old one.

7 The six empirical misses

Six of 25 empirical domains produced a best-fit model outside the predicted family (Table 3). Each miss is given an explanation below; the explanations were written after the fits and are not tested here.

In summary: three misses involve small or truncated datasets (species-area, time crystal, Zipf); one involves data that do not span the full saturation range (muscle force-velocity); one involves a classically contested relationship (Kleiber); and one involves astrophysical complexity at extreme values (stellar mass-luminosity). The explanations are post hoc, and no miss can be dismissed: whether any miss reflects its dataset rather than a wrong prediction is for the replication to decide (Section 13.1, item 4). These are genuine failures that an independent replication must address.

8 Secondary tier: published exponents

8.1 Direct published exponents (13 of 13)

Thirteen taxa with well-established metabolic scaling exponents were compared with the dimensional comparator $d/(d+1)$, the expression the derivations of Section 2 reach; it applies under the conditions of Assumption 1.

Table 3. The six empirical misses: the domains of Table 2 whose AICc-best model lies outside the predicted family, numbered as there, with the explanation given for each. Source: this paper's fits.

#	Domain	Predicted	AICc best	Explanation
1	Kleiber's Law	Power law	Saturation-exp	Classic contested case. Kleiber's original 1932 dataset has only 13 mammals. A bounded model can capture slight upper curvature at extreme body masses. The power-law exponent debate (0.67 vs 0.75) has persisted for decades.
3	Species-Area (Galapagos)	Power law	Hill	Small dataset (30 Galapagos islands) with high scatter. Several islands have very small areas with disproportionate species counts. The Hill function can absorb this scatter with its extra parameter.
6	Zipf's Law	Power law	Michaelis-Menten	Finite-corpus truncation. The Brown Corpus (Kučera and Francis 1967) has a hard upper bound on word frequency (69,971 for 'the'). At low ranks, frequencies saturate against this ceiling, pulling AICc towards a bounded model. Zipf's law is a power law in the asymptotic limit; finite corpora violate this limit.
15	Muscle Force-Velocity	Bounded	Exponential	The Hill (1938) force-velocity data show a monotonic decrease from maximum isometric force to zero at maximum velocity. The data do not reach or demonstrate an asymptotic plateau; they terminate where force reaches zero. An exponential decay fits this truncated range better than a saturation curve. The bounded prediction requires data spanning the full approach to saturation.
20	Time Crystal Order	Bounded	Power law	Only 4 data points, from a source that could not be found (Section 11.1). With four points the three-parameter saturation curves (logistic, Hill) cannot be scored by AICc, and the fitter rejected the two-parameter ones (Michaelis-Menten, saturation exponential) as invalid fits, so no bounded candidate was scored and the power law had the lowest AICc. The bounded prediction may be correct but is untestable with this dataset.
21	Stellar Mass-Luminosity	Power law	Logistic	The mass-luminosity relation shows significant scatter at high stellar masses where radiation pressure, convective instability, and different fusion pathways create deviations from a simple power law. A logistic model absorbs this scatter by fitting an apparent upper rollover.

Assumption 1 (The conditions of the dimensional comparator)**.** (a) Multiplicative composition (Theorem 1(ii) fixes the power-law family); (b) d-dimensional space-filling transport geometry; (c) a conservation or optimisation constraint on resource flow (energy minimisation in West et al. 1997; supply-demand balance in Banavar et al. 2010; steady-state energy balance in Demetrius 2003, 2006; Demetrius and Tuszynski 2010). Under (a) to (c), and when total demand scales as L^d and the demand-weighted mean route length as L, with L a dimensionless linear size, the throughput exponent against flow-length cost is, asymptotically, $d/(d+1)$ (Eastwood 2026g). Reading it as metabolic rate against body mass needs a further bridge from transport to mass: in the models of Banavar et al. (2010), the mass exponent is $2/3$ when flow velocity is independent of body size and reaches $3/4$ when velocity scales as $M^{1/12}$.

Neither Cauchy alone, nor space-filling alone, is sufficient. Where throughput is the metabolic rate and body mass M scales in proportion to flow-length cost, neither of which the geometry supplies (Eastwood 2026g), organisms with three-dimensional vascular transport and conserved flow should scale as $M^{3/4}$, those with two-dimensional transport as $M^{2/3}$, and those with one-dimensional filamentous transport as $M^{1/2}$.

Note on the $3/4$ exponent. The empirical value of the mammalian metabolic scaling exponent is debated, with estimates ranging from approximately 0.67 to 0.75 depending on taxon, mass range, temperature correction, and statistical method (White and Seymour 2003; Glazier 2005, 2022). The largest modern dataset (White and Seymour 2003, compiling 619 species) gives a maximum-likelihood estimate of 0.687 (95% CI 0.674 to 0.701), which excludes 0.750. The $d/(d+1)$ prediction of 0.750 for $d = 3$ matches the upper end of the empirical range. The variation itself is consistent with the framework: organisms with effective transport dimensions between 2 and 3 would produce exponents between $2/3$ and $3/4$. The framework accommodates this variation rather than requiring a single universal exponent; since such dimensions can produce any exponent in that range, the variation is not evidence for it.

The compendium value at row 26 is 0.737, and its interval admits 0.750, whereas the 619-species estimate given in the note above rules that value out; each figure is stated here. Every one of the 13 published exponents sits closer to its assigned comparator $d/(d+1)$ than to the comparator of any other dimension, $d = 1$, 2 or 3. Each of the 13 intervals behind the column 'CI includes?' of Table 4 contains the predicted value, but those intervals are the suite's own: its fitting code forms each as a 95% normal-theory interval around the observed exponent, from a standard error that the suite's manifest records for each row and that this paper does not anchor to a source (the provenance note below). In each of the three anchored rows whose source reports two estimates (rows 26, 29 and 30), the value printed is the one nearer 0.750; the other estimate (0.712, 0.80 and 0.82) lies nearest the same comparator, and no rule for choosing between the two is stated. Row 38 compares a different variable pair, and rows 33 to 35 take $d = 2$ without a two-dimensional transport network (below), so for those rows the match is proximity only. An interval that contains the comparator fails to reject it; that is not evidence that the exponent equals it, which would need a stated tolerance and an analysis able to resolve it (Eastwood 2026g, Appendix F). This is corroborating evidence whose per-row sourcing is outstanding for 7 of the 13 rows (see the provenance note below), but it tests a different claim (specific exponent) than the primary result (functional family), so it is reported separately.

Provenance note on the 13-taxa table. Six rows of Table 4 carry a source: those whose printed value equals the published estimate that the companion framework paper reads at its source (Eastwood 2026g, Appendix F). They are the mammals (row 26: 0.737 with mass binning, 0.712 without it; Savage et al. 2004), the reptiles (row 29: 0.76 by ordinary least squares at 38 °C, 0.80 by reduced major axis; White et al. 2006), the insects (row 30: 0.750 after phylogenetic correction, 0.82 before it; Chown et al. 2007) and the three fungal rows (36 to 38: 0.58, 0.53 and 0.53 at 20 °C; Table 1 of Aguilar-Trigueros et al.

Table 4. Direct published exponents for 13 taxa, numbered after the 25 domains of Table 2: the comparator $d/(d+1)$ for the assigned transport dimension (Predicted), the published exponent (Observed), the nearest comparator, and whether the suite's interval for it includes the prediction (the intervals are not printed or anchored to a source here; the text of this subsection says how they were formed). The source column names the published estimate for the six rows anchored at the value printed; the other seven rows are outstanding (see the provenance note in this subsection).

#	Taxon	Predicted	Observed	Nearest	CI includes?	Source
26	Mammals	0.750	0.737	$d = 3$	Yes	Savage et al. (2004), with mass binning
27	Birds	0.750	0.720	$d = 3$	Yes	outstanding
28	Fish	0.750	0.800	$d = 3$	Yes	outstanding
29	Reptiles	0.750	0.760	$d = 3$	Yes	White et al. (2006), ordinary least squares, 38 °C
30	Insects	0.750	0.750	$d = 3$	Yes	Chown et al. (2007), after phylogenetic correction
31	Amphibians	0.750	0.740	$d = 3$	Yes	outstanding
32	Crustaceans	0.750	0.730	$d = 3$	Yes	outstanding
33	Jellyfish	0.667	0.680	$d = 2$	Yes	outstanding
34	Cnidarians	0.667	0.700	$d = 2$	Yes	outstanding
35	Ctenophores	0.667	0.660	$d = 2$	Yes	outstanding
36	Ectomycorrhizal fungi	0.500	0.580	$d = 1$	Yes	Aguilar-Trigueros et al. (2017), Table 1, 20 °C
37	Marine fungi	0.500	0.530	$d = 1$	Yes	Aguilar-Trigueros et al. (2017), Table 1, 20 °C
38	Saprotrophic fungi (20 °C)	0.500	0.530	$d = 1$	Yes	Aguilar-Trigueros et al. (2017), Table 1, 20 °C

2017). Each anchored value is one of the estimates its source reports, under the method named, and two of these sources draw conclusions that bear against a transport-network account of the exponent: White et al. (2006) take the significant heterogeneity of the exponents of fish, amphibians, reptiles, birds and mammals at a common 38 °C to suggest that there is no universal metabolic allometry, and Chown et al. (2007) report that the scaling of insect metabolic rate is inconsistent with the nutrient supply network model. The three fungal fits are colony comparisons on different variable pairs: the ectomycorrhizal and marine fits relate colony respiration to colony biomass, while the saprotrophic fit relates specific metabolic rate to radial extension rate, from data the compilation credits to Crowther and Bradford (2013), so row 38 is not an exponent against biomass. The other seven rows (27, 28 and 31 to 35) carry no anchored source. The compendium literature on interspecific metabolic scaling (West et al. 1997; Banavar et al. 1999, 2010; Demetrius 2003, 2006; Demetrius and Tuszynski 2010; White and Seymour 2003; Glazier 2005, 2022; Kolokotrones et al. 2010), from which the mammalian, vertebrate and invertebrate scaling exponent literature is compiled, names the broader field, but this paper does not anchor a per-row source to those seven observed exponents or to any confidence interval in the table. Independent readers should treat those rows as pending a per-row source audit before reuse in secondary work.

Caveat on $d = 2$ biological entries. The jellyfish, cnidarian, and ctenophore entries (rows 33 to 35) are classified as nearest-comparator matches to $d = 2$, meaning their observed exponents fall closer to $2/3$ than to either $1/2$ or $3/4$. However, the $d = 2$ biological prediction remains untested in the strong sense because no known organism possesses a genuinely two-dimensional hierarchical space-filling transport network with conserved flow. These organisms lack the vascular architecture that the $d/(d+1)$ derivation assumes. The value $d = 2$ appears in cosmology only as an identity, which carries no evidence: in the

matter-era solution of Friedmann's equations for a flat universe the scale factor grows as $t^{2/3}$ (Friedmann 1922; Einstein and de Sitter 1932), and that exponent takes the form $d/(d+1)$ with $d = 2$ under a change of variables (Eastwood 2026g). In physics, the companion framework paper treats the percolation value as an analytic identity and holds the fragmentation values as unsourced (Eastwood 2026g). Neither field supplies a dimensional confirmation of $d = 2$, and biology does not either. These rows should be understood as empirical proximity to the $d = 2$ comparator, not as dimensional confirmation.

8.2 Provisional published exponents (3 of 6)

Six additional taxa with contested or proxy-based dimensional assignments were tested. Only 3 of 6 match the nearest predicted comparator. The misses are flatworms (disputed 2D vs 3D transport), bryozoans (colonial organisms with anomalous scaling near 1.0), and glass eels (elongated but retaining 3D vascular transport). These are noted for completeness but carry low evidential weight due to the contested dimensional assignments.

8.3 Analytic identities (6 of 6)

Six domains defined by exact analytic formulae ($E = mc^2$, hydrogen energy levels, Arrhenius kinetics, Michaelis-Menten kinetics, Friedmann matter-era expansion, de Sitter dark-energy expansion) were included as sanity checks. All 6 confirm. These are tautological (fitting a curve to data generated by an exact formula must recover that formula) and carry no independent evidential weight. They confirm only that the fitter works correctly.

8.4 The March 2026 extension folder, and the follow-on runs of the same era (recorded 25 August 2026)

The folder prepared in March 2026 to preregister this paper's extension is preserved in the public repository (experiments/cauchy-unification__Paper-VII/preregistration/): a written protocol; a candidate manifest committed at 00:19:19 UTC on 17 March 2026, whose own status field at that commit read `draft_local_not_locked`, carrying twelve rows each with operator class, predicted family, predicted model and target source fixed before any fit; the author's own file checksums; and the OSF component registration text, already written. Two steps were not taken: the registry submission, and the folder's own written condition that the predictions be deposited with a registry before the domain data were extracted (below). The folder's status field records the limitation in its own words: "`archived_local_packet_exercised_before_timestamp`", and the analysis is pinned to a git commit. The predictions were fixed in public before the fits: the manifest was committed to the public repository at 00:19:19 UTC on 17 March 2026, twenty-three minutes before the fits ran, and the folder, with its status field relabelled after the run and its twelve predictions unchanged, was deposited to the public OSF component between 02:06:51 and 02:07:11 that night, where it remains unaltered (Section 4.1). What is absent is an accepted registry registration, a separate schema-bound object this component does not carry; the API reports `registration: false`. The folder's status field read `draft_local_not_locked` at the commit that fixed the predictions and was rewritten to `archived_local_packet_exercised_before_timestamp` at 00:56, after the run.

Two further March 2026 runs existed only as repository artefacts until the version of 25 August 2026, and both are recorded here with their status stated before their numbers. Both ran under the pre-correction candidate set (no logarithmic candidate), and the extension's design lineage was later retired unedited. The 12-domain run is therefore exploratory support and the temporal run a dated result

under the superseded candidate set; both are superseded by the fourth-cell test, which is preregistered by timestamp (Appendix C) and awaits human submission, and neither is confirmation of anything.

The 12-domain extension of 17 March 2026 returned 10 of 12 family matches and 8 of 12 exact model matches (binomial against one third, $p = 5.4 \times 10^{-4}$; results_preregistered_extension.json), its predictions committed twenty-three minutes before the fits. Its rung: author-classified rather than externally classified; pre-correction candidate set; public repository commit attestation, with the outcome file not among those deposited to the component; and one row, EXT-10 receptor occupancy, run on Hill's 1910 haemoglobin data, repeats a domain, an operator class and a predicted model already public at 23:54 the previous evening with a matching outcome, so the strictly fresh set reads 9 of 11 ($p = 1.4 \times 10^{-3}$), or 8 of 10 ($p = 3.4 \times 10^{-3}$) if enzyme kinetics is removed with it. It is a dated result at that rung and not confirmation of this paper's claim. The demotion of that run's status is dated to the minute in the public repository's commit history: `e5c22dd` records the 10 of 12 result at 00:43 on 17 March 2026, `f5e06a8` adopts the conservative miss posture at 00:47, and `c853277` downgrades the run to a pilot dry run at 00:56, thirteen minutes after the result was recorded and months before any external review existed. The result was never retracted; its claimed status was corrected, by the author, against interest, in public. That demotion is kept here exactly as it was written. Its first ground, that a registry timestamp was required, was corrected on 6 September 2026 by the programme's attestation-class rule, fixed that day: a prediction whose text is public and timestamped before its outcome is described on those terms, with its attestation class, and is not demoted for the absence of a registry form; the record was re-read under that rule on 7 September 2026, and the commit of 00:19 had already fixed these predictions before the run. Its second ground stands: the folder's own written rule required the predictions to be deposited with a registry before the domain data were extracted, and the run and the data extraction preceded that night's deposit (Eastwood 2026m). The correction changes the status label and none of the numbers, and the run is described as timestamped in advance of its fits, never as registered or preregistered.

The seven-domain temporal out-of-sample run fitted the candidate families to older data in each domain and asked whether the family that fitted the older data best also gave the best forecast of the newer data (results_temporal_out_of_sample.json). In four domains, which its results file labels "confirmed", it did, and that family was the one predicted for the domain; in two, labelled "partial", the predicted family fitted the older data best but not the newer; and in one, labelled "inconsistent", the predicted family fitted neither best. The inconsistent domain is reported with the same prominence as the others. The criterion ranks the candidate families' forecasts against one another and does not test that any forecast was accurate: in two of the four domains labelled confirmed, the best forecast predicts the newer data less well than their own mean does, with out-of-sample R^2 of about -75.7 for neural scaling and about -0.98 for the early Wuhan wave of COVID-19 cases. A 105-domain expansion catalogue (rows 26 to 65 and 66 to 105) is prepared in the repository data and has not been run; no result beyond the runs named here is claimed.

The outermost ring of the same dated chain is in print. The first edition of 2 January 2026 (ISBN 978-1806056200; Eastwood 2026a) carries Appendix F, five numbered testable predictions closing with a printed wager: "These predictions are my wager. If they fail, the framework is wrong or incomplete. If they succeed, something important has been glimpsed. Time will judge." Appendix A operationalises the framework's propositions, states four further predictions in §A.3 (AI development quadratic rather than linear; consciousness and integrated information; cosmological recursive error correction; value embedding), and fixes falsification criteria in advance in §A.4. Prediction 2 prints the 15 per cent and 5 per cent alignment-drift thresholds that a prepared draft registration now carries unchanged into a formal hypothesis, citing the appendix by name. The book is a printed prior statement rather than a preregistration, and the two are never conflated: the print fixes the prediction; only a registration fixes an analysis plan.

9 Significance framing

The primary result ($p = 7 \times 10^{-5}$, permutation) deserves honest contextualisation.

In absolute terms: under the uniform-chance binomial null, roughly 1 in 64,000: if 64,000 researchers each picked scaling families at random for 25 domains, on average one would match 19 or more by chance. The stricter permutation null, which concedes that power laws are simply common among real fits, still leaves $p = 7 \times 10^{-5}$.

Relative to convention: the permutation test lands nearly three orders of magnitude beyond $p < 0.05$, and the binomial test more than three. Sigma equivalences are not quoted because they shift with the choice of null.

What it supports: the result is strong enough to publish, as a structured prediction comparison under a publicly stated rule rather than a per-domain advance timestamp, as evidence that Cauchy-type composition constraints predict scaling families at well-above-chance rates.

What it does not support: it is not strong enough to claim the theory is proven. The particle-physics discovery threshold is 5σ ($p \approx 3 \times 10^{-7}$). The test lacks formal preregistration. The operator classifications were made by the author, not an independent panel. The candidate set contained no logarithmic and no pure linear model, so the hit rate is conditional on that model set (Section 4.3).

What is needed: independent replication with preregistered operator classification, ideally by researchers with no stake in the outcome.

10 Group analysis

10.1 Multiplicative composition: power laws

Of 14 empirical domains classified as multiplicative, 10 produce power-law scaling as their AICc-best model. These span metabolic allometry (heart rate vs body mass), urban economics (Bettencourt et al. 2007), technology learning curves (Wright's law), computational linguistics (Heaps' law), skill acquisition (Crossman 1959), Brownian motion (Catipovic et al. 2013), neural network scaling (Kaplan et al. 2020), VLSI design (Rent's rule), population ecology (Taylor's power law), and fluvial geomorphology (Hack's law).

Four multiplicative domains miss: Kleiber's law (bounded model edges out power law on a 13-point dataset), species-area (high scatter on small island data), stellar mass-luminosity (scatter at high mass), and Zipf's law (finite-corpus truncation).

10.2 Additive composition: exponentials

All four domains with additive composition operators produce exponential scaling as their AICc-best model: Moore's law (transistor counts), radioactive decay (P-32), the Gutenberg-Richter earthquake frequency law, and Horton's law (stream numbers). This is 4 of 4; two of the four use reference or compiled values (radioactive decay and Moore's law; Section 11.1).

10.3 Bounded composition: saturation curves

Of seven domains with bounded composition operators, five produce bounded (saturation) models as AICc-best: bacterial growth (logistic), haemoglobin binding (Hill), Ebola epidemic (Hill), Amdahl's law (Michaelis-Menten, with values computed from its own formula; Section 11.1), and Facebook growth

(logistic). Two miss: muscle force-velocity (exponential decay fits the truncated range better) and time crystal order (only 4 data points).

11 Data provenance

11.1 Provenance caveat

Three domains use theoretical or reference values rather than independent empirical measurements: radioactive decay (values computed from a reference half-life for P-32 of 14.29 days, whose printed publisher and value could not be confirmed at an official source; its entry in the references, cited in Appendix E, gives the current evaluated value), Amdahl's law (the theoretical formula with an assumed 10% serial fraction, itself of the Michaelis-Menten form that was selected), and Moore's law (transistor counts compiled on Wikipedia from manufacturer specifications). The first two are analytic identities in the sense of Section 8.3, counted among the 19 matches; all three match trivially and should be weighted accordingly. One further domain has a data source that could not be found: the time-crystal domain (domain 20) credits its four points to a 2025 paper by Shen and colleagues in *Nature Communications*, of which no record was found at Crossref or at the journal on 1 November 2026 (Shen et al. 2025), so the provenance of its data is unconfirmed; it is one of the six misses.

11.2 Citations for all 25 empirical domains

The source of each of the 25 empirical domains of Table 2, with the data notes of version 3.15 as printed, is listed in Table 10 of Appendix E. The works cited there, the compendium literature on metabolic scaling behind Table 4 and the other works cited in the text are in the references.

12 What this paper does and does not show

12.1 What it shows

As mathematics: under a stated regularity condition, the non-zero solutions of Cauchy's four functional equations form exactly four families, linear, power law, exponential and logarithmic (Theorem 1). Read as composition rules in a declared variable, they give a power law where depth and output both multiply and an exponential where depth adds and output multiplies (Definition 1 and Remark 2), and a non-constant response that tends to a finite non-zero ceiling satisfies none of them in that variable (Proposition 1).

As evidence: a structured prediction comparison under a publicly stated rule. The composition operator of each of 25 empirical domains was classified from known physics before the fitting procedure ran, and the predicted family was the family that AICc model selection returned in 19 of the 25 (marginal-conditioned permutation $p = 7 \times 10^{-5}$), on the six-model candidate set of Section 4.3, and in 14 of the 18 multiplicative and additive domains (permutation $p = 3.0 \times 10^{-4}$; Section 6). The secondary tiers are reported separately and never blended (Sections 5 and 8).

12.2 What it does not show

- **It is not proven.** The comparison's predictions were not timestamped in advance of their own fits, its operators were classified by the author, without blinding to the forms established for most of its domains, and its candidate set had no logarithmic and no pure linear candidate (Sections 4.1 and 4.3). A replication with independent classification and a four-family test are preregistered by timestamp (12 August 2026, Bitcoin block 962066) and await human submission.

- **No registered verdict on P18 or P9.** This paper informs Law I through two propositions of the theory registration, P18, that the composition operator predicts the family, and P9, the cross-domain form (Eastwood 2026m). Its comparisons are dated results under the rule of March 2026, and the registration scores none of them against any prediction.
- **No universal classification of nature.** Theorem 1 classifies the solutions of four functional equations. It does not exhaust scaling laws, because most systems are not homomorphisms of either structure (Section 3.1), and finitely many noisy observations cannot establish that a domain's composition rule holds exactly, or within an error bound over the whole domain (Remark 3).
- **No exponent selected by Theorem 1.** The theorem fixes a family and leaves its constant c free. The comparator $d/(d+1)$ of Section 8.1 rests on the transport conditions of Assumption 1, not on the theorem.
- **The bounded class rests on added assumptions.** No saturating form follows from the four equations (Proposition 1): the bounded class rests on a ceiling on output and on dynamics of its own, and a ceiling alone selects no form (Section 3.3).

12.3 The programme's status

The theory registration (DOI 10.17605/OSF.IO/P8CKQ) was registered on 8 September 2026 at version 1.100 and updated on 13 September 2026 (1.102), 15 October 2026 (1.103) and 29 October 2026 (1.104, approved by OSF that day, the current version); 1.100, 1.102 and 1.103 stay frozen beside it. Laws I to III were registered on 8 September 2026, and Laws IV and V, the ARC Persistence Law and the ARC Embedding Law, as laws on 15 October 2026 in version 1.103. Each is a law under test. Law IV is not yet tested. Law V is under test: at the level of its proposition P21, "P21 is supported under the registered test, by the programme's own run, on the access axis, family-dependent, with its mechanism axis not evaluable and one call retried under a disclosed recovery amendment" (Unit 83, registration DOI 10.17605/OSF.IO/MN73W and results DOI 10.17605/OSF.IO/98HKR; Eastwood 2026p,q); at the level of the law, the registration's verdict rule prints "supported on the performance axis only . . . which is not support for the information-asymmetry mechanism", and its build-order limb, read on P12, is undecided. Their mathematics is in development (version 1.2, 29 October 2026, DOI 10.17605/OSF.IO/4B3VT).

The first registered study, Unit 06 (a classification audit, DOI 10.17605/OSF.IO/RV2B5), was registered on 1 October 2026 and ran on 6 October 2026; it is reported in full with its five deviations, as administratively incomplete, with no inferential verdict (results DOI 10.17605/OSF.IO/4BPHG), and it tests no law.

This paper's results decide none of P15, P12 and P21 and test neither Law IV nor Law V; naming the laws changes nothing printed in its results.

Epistemic status. What this programme names Laws are conjectures, each a law under test; the one registered verdict so far is Unit 83's on P21, Law V's placement limb. The programme's symbols are operationally defined in its Operational Definitions (Eastwood 2026f), and established-law standing is claimed nowhere. The registered programme exists to earn that standing, or lose it, by measurement, replication and survived refutation.

13 Limitations

- **No registry submission for this test, and no advance per-domain timestamp for this cohort.** The predictions, data arrays, and fitting logic all reside in a single author-written script. The timing of the classification rule, the per-domain assignments and the fits is set out in Section 4.1.

- **Operator classification by author.** The composition operator classification is done by the author, not by an independent assessor. A replication with independent operator classification would be more rigorous.
- **The classification was not blind to the established forms.** Most of the 25 domains are named laws whose functional forms are established in their fields (Kleiber's, Zipf's, Heaps', Moore's, Wright's, Rent's, Taylor's, Horton's and Hack's laws among them), and their operators were classified by the author with those forms already published. A classification made before the fit is therefore not blind to the expected form, and the comparison cannot separate what a domain's composition operator predicts from what is already known of its form. The replication with independent classification (Section 4.1) separates them only if its classifiers do not know each domain's established form, or if its domains have none.
- **The bounded class rests on added assumptions, and a bound selects no form.** No saturating form follows from the four equations: each bounded candidate rests on a ceiling on output, an added assumption, and satisfies one of the equations only in a variable that the ceiling defines (Proposition 1); which curve a bounded domain follows is fixed by dynamics of its own (Section 3.3). The bounded category groups the logistic, Hill, Michaelis-Menten and saturation-exponential forms (with the hyperbolic decay for domain 15, Section 4) by empirical convention rather than mathematical proof.
- **Legacy AIC was an approximation.** The v1 test used an R^2-based AIC approximation, equation (1), not a full likelihood-based nonlinear AIC. The v2 test corrects this with proper AICc computed from residual sum of squares.
- **Two candidates are fitted on another scale.** The power law and the exponential are fitted on the logarithm of the response and scored by AICc on the response itself, while the saturation curves are fitted and scored on the response (Section 4.2). A re-run that fits every candidate on the scale AICc scores is needed to show whether any selection depends on it.
- **Three datasets use reference values.** Radioactive decay (a reference half-life whose publisher could not be confirmed), Amdahl's law (its theoretical formula) and Moore's law (compiled manufacturer specifications) use theoretical or reference values rather than independently collected empirical measurements; the first two are analytic identities counted among the 19 matches, and the data source of the time-crystal domain could not be found (Section 11.1).
- **Family prediction only.** The framework predicts the functional family but not the specific exponent. Predicting exponents requires additional structure (dimensionality, constraint type) developed in the companion Foundational paper (Eastwood 2026d).
- **Six empirical misses.** The misses are not dismissible. They identify real limits of the framework: small datasets, truncated ranges, high-scatter regimes, and domains where competing physical effects (radiation pressure, corpus boundaries) perturb the ideal prediction.
- **Operator classification is descriptive, not explanatory.** The framework does not explain *why* a particular domain has a multiplicative rather than additive composition operator. It takes the operator as given and predicts the consequence.
- **Baseline-20 rerun is not cleanly comparable to v1.** The fitter changed from R^2-based AIC approximation to proper AICc. The model set expanded. A direct comparison of 14/20 (v1) to 15/20 (v2 baseline rerun) is not meaningful because the selection criterion changed.
- **Publication bias risk.** The 5 new expansion domains were selected after the baseline-20 test. While they were chosen to span new physical regimes (stellar astrophysics, allometric heart rate, VLSI, ecology, geomorphology), the selection was not randomised.

- **Finite model set, and specifically two Cauchy families out, one non-Cauchy family in.** The fitter compares six candidate models (power law, exponential, logistic, Hill, Michaelis-Menten, saturation-exponential). Two of the four Cauchy solutions named in Section 3.1 have no candidate (logarithmic $c \log x$; pure linear cx), while the bounded family, which rests on added assumptions (Section 3.3), is represented four times. The true best model for any domain may not be in this set, and Section 4.3 states the consequences for the hit counts.
- **Sample size.** Twenty-five empirical domains is a reasonable first sample but not definitive. Expansion to 100+ domains with preregistered selection criteria would strengthen or weaken the result.

13.1 Falsifiability: what would defeat this claim

The limitations above describe where the framework is weak; this subsection states the specific outcomes that would falsify it. The central claim, that a domain's composition operator, classified from known physics *before* fitting, predicts its scaling-law family, is defeated by any of the following:

1. **Hit rate at chance.** The primary result is a family-prediction hit rate of 19/25 empirical domains (Sections 6 and 7), on the six-model candidate set of Section 4.3. With three to four candidate families, assignment at random gives a chance baseline of roughly one in three to one in four. The claim is defeated if a replication whose domain list and per-domain predictions are timestamped in advance, and whose operator classes are assigned by an external human classifier, returns a hit rate not significantly above that chance baseline. The advance-timestamped run on the record is the 12-domain extension of 17 March 2026, its predictions committed to the public repository at 00:19:19 UTC in commit 8edaec0, attestation class public repository commit date, its fits run twenty-three minutes later: 10 of 12, and 9 of 11 on the strictly fresh rows. Its predictions were thus fixed and timestamped in advance of its fits; it is described neither as registered nor as preregistered, and it is not an accepted registry submission; its twelve classifications are the author's own, so this criterion has not yet been put at risk in the form stated here.
2. **Post-hoc operator classification.** The test's strength depends on the operator being assignable from independently known physics without seeing the data. The claim is defeated if, in a blind trial, independent assessors cannot reproduce the author's operator classifications from the physics alone, or if any classification can be shown to have been chosen to fit the observed exponent.
3. **Fit degeneracy.** The prediction is meaningful only if the candidate families are empirically distinguishable on the data at hand. It is defeated for any domain where the fitter reaches the observed curve equally well under the *wrong* family; that is, the family assignment carries no discriminating information for that domain's range and noise.
4. **Unrationalised misses.** Six domains miss (Section 7). The misses are consistent with the claim only if their causes are fixed as a preregistered exclusion rule *before* the next test, not supplied after seeing which domains missed. The claim is defeated if the miss rate rises under a preregistered replication with no post-hoc exclusions.
5. **Four-family re-run.** The four-family confirmatory study, preregistered by timestamp (Section 4.3), re-runs this comparison with logarithmic and pure linear candidates included. The claim is weakened if the hit rate falls materially when the missing families are available, and defeated if it falls to the chance baseline; hits that migrate to the logarithmic family will be reported as reclassifications, not as new confirmations.
6. **Exploratory ceiling.** Until the operator classes and the per-domain predictions for a randomised domain sample are fixed before any fitting by an external human classifier and timestamped

in advance with the attestation class stated, the claim cannot rise above *exploratory*, and any presentation of it as established would itself falsify the paper's stated evidential status. A registry deposit raises the attestation class; it is not what lifts the ceiling. Part of that condition is already met on the record: the 12-domain extension of 17 March 2026 (item 1) had its per-domain predictions committed before its fits; its classifier was the author, its candidate set was the pre-correction one, and the author's demotion of 00:56 that night stands so far as it rests on the packet's own written condition, deposit to a registry before data extraction, which was not met, and is reversed as of 7 September 2026 only so far as it rested on the absence of a registry form.

What would *not* defeat it: a single additional miss, or a better-fitting model appearing in one domain. The claim is about the aggregate operator-to-family mapping, not any single fit; but that aggregate must clear the chance baseline under preregistered, independently classified replication.

Standing covenant. Prove this paper wrong, and I will publish the refutation myself. Falsification conditions are stated in this paper; the standing challenge:
github.com/MichaelDariusEastwood/arc-scaling-challenge.

14 Implications

If the framework holds under independent preregistered replication, it would suggest:

1. The forms of scaling laws are not coincidences but mathematical consequences of composition structure: a power law where composition is multiplicative and an exponential where it is additive, in the declared variable and under the regularity of Theorem 1, while a saturating form needs in addition a ceiling on output and dynamics of its own (Proposition 1).
2. The same composition rule that fixes the power-law family of a mouse's heart rate would fix the family, though not the exponent, of how neural networks improve with scale.
3. Critical phenomena would be one case of the wider rule: scale invariance gives their power-law form through the multiplicative equation, while their exponents are left to the renormalisation group. Section 2 sets out what earlier work already states on scale invariance, universality and renormalisation (Newman 2005; Stanley 1999), and what this paper adds.
4. Any new system's scaling family can be predicted from first principles by classifying its composition operator, before any data fitting. The prediction follows only where the composition rule holds exactly, or within an error bound over the whole domain, in a declared variable (Remark 3); finitely many noisy observations cannot establish such a bound, so each such prediction remains a hypothesis for the system's data to test.
5. Domain-specific explanations of the forms of scaling laws (fractal networks, learning curves) may share a single constraint on form, that of the functional equations Cauchy (1821) solved, while the values of their exponents, and behaviour outside the four families, need physics of their own (Section 3.1).

These implications are conditional. The current evidence is not definitive.

15 Conclusion

Plant an acorn and given centuries, you get an oak. But plant that oak's acorn, and its acorn, recursively across millennia, and you get a forest that shapes the climate of continents. What we plant in these systems

will compound across scales we cannot imagine. *The seed determines the forest.*

Read as composition rules, the functional equations Cauchy solved 200 years ago (Cauchy 1821) say that the seed also determines the *form* of the forest. Multiplicative seeds produce power laws. Additive seeds produce exponentials. Both hold under the regularity of Theorem 1, in the variable in which depth composes. Bounded seeds can produce saturation, but not from Cauchy's equations: the ceiling is an added assumption (Proposition 1), and the form of the saturation needs dynamics of its own. Twenty-five empirical domains tested against this prediction yield 19 matches at $p = 7 \times 10^{-5}$ under the marginal-conditioned permutation test, on the six-model candidate set of Section 4.3; the four-family re-run is preregistered by timestamp and awaits human submission. Thirteen published metabolic exponents sit nearest the comparator the dimensional theory predicts.

The mathematics was always there, and earlier work had read invariance as fixing the possible forms of laws (Section 2); what this paper adds is a forward test, across domains, of the form each domain's composition rule predicts. Whether that reading holds under independent scrutiny is now a question for the scientific community to answer.

Raise AI with care.

Code and data. The complete validation suite, including all 50 domain definitions, the canonical manifest and the AICc-based fitting code, is in the public repository github.com/MichaelDariusEastwood/arc-principle-validation, folder experiments/cauchy-unification__Paper-VII; the 50-domain manifest is data/canonical_50_domain_manifest.json. The script, scripts/arc_50_domain_universal_test.py, requires Python 3.10+, numpy, and scipy. The folder, the manifest and the script are linked at commit 5435e77 of 16 March 2026, which carries the 19 of 25; the manifest and the script are unchanged at the repository's main of 7 November 2026. Each domain's data are embedded in the manifest with the source it records; Appendix E gives each domain's sources, and their entries in the references say where a source could not be confirmed (Section 11.1). Output is deterministic.

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A Proofs

This appendix proves Theorem 1 and Proposition 1 and works the change of variable of Remark 2. The results of Theorem 1 are classical (Aczél 1966, chapter 2), and the framework paper proves the continuous case (Eastwood 2026g); the arguments are given in full so that each regularity condition can be checked where it is used. Logarithms are natural throughout.

A.1 Cauchy's four equations under regularity

Proof of Theorem 1. The additive equation. Let $H : \mathbb{R} \to \mathbb{R}$ satisfy $H(x + y) = H(x) + H(y)$ for all real x and y. Then $H(0) = 0$ and $H(-x) = -H(x)$; repeated addition gives $H(mx) = mH(x)$ for every integer m, and hence $H(qx) = qH(x)$ for every rational q. With $c = H(1)$, the function $G(x) = H(x) - cx$ is additive and vanishes at every rational number. Each condition of the theorem forces $G \equiv 0$.

(a) If H is continuous, so is G, and a continuous function that vanishes on the dense set of rationals vanishes everywhere.

(b) Suppose that $H \le M$ on an interval $(u, u + \delta)$ with $\delta > 0$. Then $G \le M' = M + |c|(|u| + \delta)$ on that interval. For any real t, the interval $(u - t, u + \delta - t)$ contains a rational q, and $G(t) = G(t + q) - G(q) = G(t + q) \le M'$, because $t + q$ lies in $(u, u + \delta)$; so $G \le M'$ everywhere. For every positive integer m, $mG(t) = G(mt) \le M'$, so $G(t) \le 0$; applied to $-t$, the same bound gives $G(t) \ge 0$. A bound below is the case of $-H$, and a function bounded on an interval is bounded above there.

(c) Suppose that H is Lebesgue measurable. The measurable sets $\{x : |H(x)| \le m\}$, for $m = 1, 2, \ldots$, cover the line, so one of them, E, has positive measure, and its difference set $\{x - y : x, y \in E\}$ contains an interval $(-\eta, \eta)$ with $\eta > 0$ (Steinhaus 1920). Each t in that interval is $x - y$ with $x, y \in E$, so $|H(t)| = |H(x) - H(y)| \le 2m$. Thus H is bounded on $(-\eta, \eta)$, and (b) applies.

So $H(x) = cx$, which is (i); conversely, every cx is additive.

The other three equations. Each reduces to the additive equation by a substitution that keeps the regularity conditions: exp and log are continuous, and, as locally Lipschitz bijections, they carry Lebesgue measurable sets to Lebesgue measurable sets. For (ii), $H(t) = \log f(e^t)$ is additive on $\mathbb{R}$, because $H(s + t) = \log f(e^s e^t) = H(s) + H(t)$; so $H(t) = ct$ and $f(x) = e^{H(\log x)} = x^c$. For (iii), $H(t) = \log f(t)$ is additive, so $f(x) = e^{cx} = a^x$ with $a = e^c > 0$. For (iv), $H(t) = f(e^t)$ is additive, so $f(x) = H(\log x) = c \log x$. A bound on f over an interval passes to H over the corresponding interval: as a bound on both sides in (iv), and in (ii) and (iii), where $0 < f \le M$, as the bound $H \le \log M$, which is all that (b) needs. Direct substitution shows that each stated function satisfies its equation.

Real targets. If $f : (0, \infty) \to \mathbb{R}$ satisfies $f(xy) = f(x)f(y)$, then $f(x) = f(\sqrt{x})^2 \ge 0$, and a zero at x_0 gives $f(x) = f(x_0)f(x/x_0) = 0$ for every x. If $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x + y) = f(x)f(y)$, then $f(x) = f(x/2)^2 \ge 0$, and a zero at x_0 gives $f(x) = f(x_0)f(x - x_0) = 0$ for every x. A real-valued solution of (ii) or (iii) is therefore identically zero or positive everywhere, and in the second case the argument above applies.

Positive arguments in (i) and (iii). Let $f : (0, \infty) \to \mathbb{R}$ satisfy $f(x + y) = f(x) + f(y)$ for all $x, y > 0$. For real x, put $F(x) = f(x + t) - f(t)$ with any $t > \max(0, -x)$; the value does not depend on t, since $f(x + t) + f(t') = f(x + t + t') = f(x + t') + f(t)$ for two such t and t'. Then F is additive on $\mathbb{R}$ and equals f on $(0, \infty)$. On each interval $(-N, N)$ it is the function $x \mapsto f(x + N) - f(N)$, so it is continuous or Lebesgue measurable when f is, and it is bounded on any interval in $(0, \infty)$ on which f is bounded. Hence $F(x) = cx$, and $f(x) = cx$. For (iii) with positive values on $(0, \infty)$, the same argument applies to $\log f$. With real values on $(0, \infty)$, $f(x) = f(x/2)^2 \ge 0$; a zero at x_0 gives zeros at $x_0/2^j$ for every positive integer j, and then $f(x) = f(x_0/2^j)f(x - x_0/2^j) = 0$ whenever $x > x_0/2^j$, so $f \equiv 0$. The families are therefore the same. □

A.2 Saturating responses lie outside the four equations

Proof of Proposition 1. By Theorem 1, a non-constant solution is cx or $c\log x$ with $c \neq 0$, x^c with $c \neq 0$, or a^x with $a \neq 1$; the further solution $f \equiv 0$ of (ii) and (iii) with real targets is constant. As $x \to \infty$, cx and $c\log x$ tend to $+\infty$ or to $-\infty$ with the sign of c; x^c tends to $+\infty$ if $c > 0$ and to 0 if $c < 0$; and a^x tends to $+\infty$ if $a > 1$ and to 0 if $0 < a < 1$. This proves the first statement.

For the second, suppose that a non-constant U tends to a finite $K \neq 0$ as $R \to \infty$ and satisfies one of the four equations in R. Then $|U(R) - K| < 1$ for every R beyond some R_1, so U is bounded on an interval of positive length and meets the regularity of Theorem 1 with no further assumption; for (i) and (iii) on the positive half-line, the theorem's last sentence applies. By the first statement, $U(R)$ then tends to $+\infty$, to $-\infty$ or to 0, which contradicts $U(R) \to K$. Dividing U by a positive constant, as the normalised rules of Definition 1 do, keeps its limit finite and non-zero, so the conclusion holds for those rules too.

For the third, direct algebra gives the three identities. For the saturation exponential, $K - U = Ke^{-cR}$, so $(K-U)/K = e^{-cR}$ satisfies the exponential equation in R. For the logistic, $K - U = Ke^{-\lambda(R-R_0)}/[1 + e^{-\lambda(R-R_0)}]$, so $U/(K-U) = e^{\lambda(R-R_0)}$ and $\log[U/(K-U)] = \lambda(R - R_0)$, which satisfies the additive equation in $R - R_0$. For the Hill form, $K - U = Kk^n/(k^n + R^n)$, so $U/(K-U) = (R/k)^n$ and $\log[U/(K-U)] = n\log(R/k)$, which satisfies the logarithmic equation in R/k; $n = 1$ gives the Michaelis-Menten form. Each transformed quantity is formed with K, and, for the logistic and Hill forms, with R_0 or k; none of these is fixed by a composition rule in R. □

A.3 The change of variable

Let $V(R) > 0$ satisfy $V(R_1R_2) = V(R_1)V(R_2)$ for all $R_1, R_2 > 0$, and put $W(s) = V(e^s)$ for real s. Then $W(s_1 + s_2) = V(e^{s_1}e^{s_2}) = V(e^{s_1})V(e^{s_2}) = W(s_1)W(s_2)$, so W satisfies the exponential equation, and $V(R) = R^c$ corresponds to $W(s) = e^{cs}$: a power law in R is an exponential in $s = \log R$. Conversely, an exponential W gives the power law $V(R) = W(\log R)$. Each regularity condition passes between V and W, as in Appendix A.1, so the two descriptions are equivalent, and only the declared variable decides which label applies. For the normalisation, $Y(x) = Ax^c$ with $A > 0$ gives $Y(x)Y(y)/A = A(xy)^c = Y(xy)$, so $Y/Y(1) = x^c$ satisfies the multiplicative equation, while Y itself satisfies it only when $A = 1$. In the same way, $U(R) = U(0)a^R$ satisfies the exponential equation only after division by $U(0)$, which is the normalisation of Definition 1.

B Version history and corrections

Every version and correction of this paper, with its date. Appendix B.1 gives this version's entry. The notes of earlier versions follow as they were printed, and Table 5 maps the layout they describe to this edition; Appendix B.3 lists every deposit of the paper on OSF, and Appendix B.4 records the original artwork of Figure 1.

B.1 Version 3.16

Version 3.16, 7 November 2026. This version sets the paper in the programme's presentation standard and carries the corrections owed to it; no registered prediction, kill condition or dependency changes, no result is added, and no date of any claim moves.

- The paper's displayed title is now its established name followed by its question, *Cauchy Unification: does the composition operator of a process determine the form of its scaling law?*; the title recorded on OSF stands in the first-page footnote, which also states the paper's place in the programme: it informs Law I, the ARC Principle, a law under test, through propositions P18 and P9 of the theory

registration, and decides no proposition of Laws II to V. The abstract opens with the question; an introduction gives the question, the contributions and a map of the paper; related work moves from Section 11 to Section 2; Section 12 states what the paper does and does not show; Appendices A to F give the proofs, this history, the priority record, the notation, the data sources of the 25 empirical domains and the paper's place in the programme.

- Section 3 states the four equations with their domains and the regularity under which each has one family of non-zero solutions (Theorem 1), the declared variable on which a family label depends (Remark 2), the exact composition rules (Definition 1), what an approximate rule needs (Remark 3), and that a saturating response satisfies none of the four equations, so the bounded class rests on an added ceiling (Proposition 1), with proofs in Appendix A. Each place that states the bounded class now says so, and Section 4.3 counts the candidate set as two of the four families and four saturation curves. Until this version, the abstract said that the four equations are the only continuous solutions to their respective composition constraints, Section 2.1 that they have unique continuous solutions, and the box headed 'The prediction' that the prediction follows from Cauchy's equations with no free parameters and no curve fitting.
- The count from the rerun of 11 August 2026 is no longer printed: the registration records that the author did not request that analysis and has not examined its outputs. The 19 of 25 stands with its six-model qualifier, and the dated notes in Appendix B.2 withhold that count where two of them printed it.
- The model fitting is no longer called independent: the author ran it after the classification rule was public.
- The 19 matches are no longer called confirmations: the predicted family is the one that was selected. The thirteen published metabolic exponents are said to sit nearest the comparator, where the conclusion said that they fall where the dimensional theory predicts.
- Section 8.1 states the premises under which the comparator $d/(d+1)$ holds (Assumption 1), anchors six rows of Table 4 to their sources and marks the other seven as outstanding, adds that an interval containing the comparator does not show that the exponent equals it, and states that the value $d = 2$ appears in cosmology and in percolation only as identities, which carry no evidence. Until this version, it said that under all three conditions the exponent is constrained to $d/(d+1)$, and that the $d = 2$ confirmation exists in cosmology and physics.
- Section 2 states exactly what earlier work states, including work on functional equations and the forms of laws, on scale invariance and on composition rules that version 3.15 did not cite; it names each derivation of $d/(d+1)$ in its own terms without calling any of them independent, states the account of their common form as this paper's hypothesis, and dates the literature search by the paper's public prior-art record. Section 14 no longer says that a connection between the renormalisation group's power-law predictions and Cauchy's multiplicative equation had not, to the author's knowledge, been formally articulated; it points to the earlier work conceded in Section 2.
- Section 8.4 describes the temporal test's criterion as a ranking of forecasts and names the two domains labelled confirmed in which the best forecast predicts the newer data less well than their own mean; it calls the 12-domain run exploratory support and the temporal run a dated result under the superseded candidate set, where it called both dated exploratory results.
- The status of the five laws, the registration's current version (1.104), Unit 83's result on P21 and the mathematics of Laws IV and V (version 1.2) are brought up to date in Section 12.3, which also restates the epistemic-status line; the note of 3 October 2026 is kept as printed in Appendix B.2.
- Figure 1 is redrawn. Its two left cards give the error-decay exponents that Paper II reports: for sequential recursion, one model's estimates by regression and by endpoint, with the endpoint's

interval, where the artwork paired the regression estimate with that interval and its strip of measured values named six models; for parallel recursion, the exponents without the claim of compute matching that the figure's description made. The label that read capability as measured by accuracy or error is not drawn, because an error-decay exponent is not the capability exponent without a stated metric bridge. The coupling carries its subscript, β_L, and is to be fitted from data, where the artwork said measured from data. The ARC Bound is named as the conjectured value of Law III, the ARC Ceiling, where the artwork labelled it a stable limit without naming its law. The caption now states the model and the conditions within which the self-reference theorem gives $\alpha = 1/(1 - \beta_L)$. The strip of measured values, with four-layer blinding, an error suppression of about 57 per cent per depth, a coefficient of determination of 1.000 and thirty ODE solutions, which this paper does not source, and the source line, which named a website file, are not drawn. Appendix B.4 records the earlier artwork's labels.

- Where version 3.15 printed the same account in several places, it is now given once and the other places point to it: the timing of the predictions in Section 4.1; the account of the 12-domain extension in Section 8.4 and item 1 of Section 13.1; and the R^2-based approximation of the AIC used by the first test, as Equation (1) in Section 4.2.
- Section 3.3, the abstract and each place that states the bounded class now say that a bound selects no form: a bound and convergence are different premises, and a saturation curve rests on a ceiling on output and on dynamics of its own. The list of bounded candidates names the saturation exponential, which the candidate set holds, where it named a hyperbolic form. Until this version, the abstract said that where output is bounded the compatible forms are saturation curves, and that a formal proof that they exhaust the bounded case is not provided and remains an open problem; Section 2.2 said the same, and Section 12 that bounded composition exhaustiveness is open.
- Section 4.1 no longer says that the outcomes for the domains in the deposit of 17 March 2026 had not been computed, or that the deposit shows the predictions public and unaltered before any result they predict existed: the deposit followed the twelve-domain extension's run and its relabelling, and it corroborates the domain list and its date, not the run's prospective standing, as the registration states. Section 8.4 and item 1 of Section 13.1 no longer call that extension preregistered in substance or its folder complete, and no longer say that its relabelling was reversed: the first ground of the relabelling was corrected and the second, the folder's own condition that the predictions be deposited with a registry before the data were extracted, stands, so the run is described as timestamped in advance of its fits.
- Section 4 states how the candidates were fitted: the power law and the exponential by linear least squares on the logarithm of the response, then scored by AICc on the response itself, and, for domain 15, a seventh candidate, the hyperbolic decay. Sections 4.2 and 13 state what that can and cannot change. Until this version, step 4 said that every candidate was fitted by nonlinear least-squares regression.
- Section 8.1 states that the intervals behind its column 'CI includes?' are the suite's own, from standard errors this paper does not anchor, that in each anchored row with two published estimates the one printed is the nearer to 0.750, and that rows 33 to 35 and 38 match by proximity only. It names the comparator $d/(d + 1)$ as the expression earlier derivations reach, where it called it the ARC/Cauchy dimensional prediction, and it says that the framework accommodates the variation of the exponent between taxa, where it said that the framework predicts it.
- Sections 6, 10 and 11 state that two of the 19 matches, radioactive decay and Amdahl's law, use data computed from the formula itself. Section 11.1 no longer names NIST as the source of the half-life of P-32, and it states that the data source of the time-crystal domain could not be found; Section 7

corrects that domain's explanation, since with four points no bounded candidate was scored, and it says that the explanations of the misses were written after the fits and are not tested. Sections 12 and 13 add that the classification was not blind to the forms established for most of the domains.

- Section 4.2 says that the script of the permutation test is held in a private repository and is not yet public, and that the test can be recomputed from Table 2, where it named the script without saying so. The paragraph headed Code and data links the suite's folder, manifest and script at the commit of 16 March 2026 that carries the 19 of 25, where it linked the repository's main branch, and it no longer says that the manifest carries full provenance citations.
- Sections 1 and 2 state the novelty as narrowly as the related work requires, and Section 2 adds dimensional analysis and similarity (Buckingham 1914; Barenblatt 1996) and says that nothing rests on the He and Chen item. The conclusion no longer says that Cauchy proved that the seed determines the form of the forest, or that the mathematics had not been read as a prediction about the physical world. Section 14 no longer says that one principle governs a mouse's heart and the improvement of neural networks, that a formal proof of an equivalence between the renormalisation group's power-law predictions and Cauchy's multiplicative equation remains open, or that two hundred years of explanations may be special cases of one constraint that Cauchy identified.
- Section 3.1 says that the increments of $c \log x$ diminish while the map is unbounded, so that it is not a saturating family, where it said that it is not a diminishing-returns family. Where the four-family test, the fourth-cell test and the replication were called drafted, they are now called preregistered by timestamp, as the abstract and Section 4.1 already called them. Section 9 no longer says that both tests land three orders of magnitude beyond $p < 0.05$; the 50 domains are no longer called real-world systems, since six are analytic identities; and two internal names are no longer printed.
- Section 3.1 no longer lists critical exponents among what lies outside the grid: a critical power law can have the power-law form, and Theorem 1 neither excludes it nor derives its critical exponent or physical mechanism.
- The references are set by author and year, and each links its official source or says that none could be confirmed; Appendix E gives the data source of each of the 25 empirical domains with the data notes of version 3.15. The entry for the half-life of P-32 records that neither its publisher, as printed, nor its value could be confirmed at an official source, and gives the current evaluated value; three entries whose printed details match no record say so. Section 4.3 links the dated record of the four-family test by its OSF DOI, where it linked a website page.
- The paragraph headed Code and data names the suite's script, and the copyright line takes the programme's standard form. British spelling (haemoglobin) and the eponym of Heaps' law are corrected, 'preregistered' is written as one word, and ranges are written with 'to'.

B.2 The notes of versions 3.9 to 3.15, as printed

Version 3.15 printed the following notes beneath its title, newest first, and the note of 3 October 2026 after its opening box. They are kept as they were printed, with one exception: the count from the rerun of 11 August 2026, which two of them printed, is withheld and marked as withheld in square brackets, because the author did not request that analysis and has not examined its outputs (Section 4.3). Their other counts and statuses are those of their dates, among them the status of Laws IV and V before they were registered; Sections 4.3 and 12.3 give the current statements. The sections, boxes and tables they name are those of the layout of version 3.15 and earlier, which Table 5 maps to this edition; the references they number are listed with those numbers in Appendix E.

5 October 2026 (v3.15): the title now follows the programme's uniform order (paper, its law or framework, its own title) and reads *Paper VII · The ARC Theory · Law I, the ARC Principle · Cauchy Unification: the Classification Behind the Law's Form*; until this version it read *Cauchy Unification: the Classification Behind Law I's Form (Paper VII)*. The plate beneath the author line now gives the paper's place in the programme in the words of the programme's map. Mentions of the programme's papers in this paper now carry their uniform titles. The opening box now names Law I, the ARC Principle, as the law whose form the paper classifies. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

4 October 2026 (v3.14): the title now names the law whose form the paper classifies, Law I; until this version it was titled *Paper VII: Cauchy Unification*. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

4 October 2026 (v3.13): Section 11 no longer says that the $d/(d+1)$ formula was independently derived by at least seven research groups; as the works read, one is a special case, one could not be confirmed at a registry and the abstracts of two state no exponent formula, so a count of independent derivations granted earlier work more than it stated. It now says that the formula is derived, in its general form or as a special case, in earlier work, and its closing clause speaks of these derivations without calling them independent. The version number and revision date in the header, the title block, the summary, the structured and citation records and the footer. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

3 October 2026 (v3.12): the theory gained two laws on 24 September 2026, and a dated note after the opening box now places this paper against Laws IV and V: its results do not test either law. The abstract said that Cauchy's result constrains scaling laws to one of three functional families, saturation among them; it now says what Section 2.2 states, a power law for multiplicative composition and an exponential for additive composition, with saturation curves compatible with bounded composition and the proof that they exhaust that case still open. The summary above the paper and the opening box said the same of the three families and now say it in those terms. The conclusion counted bounded seeds among those whose form Cauchy proved; it now says that this case is not strictly Cauchy-derived and that its exhaustiveness remains open. Section 1 said that the paper proposes an answer in which the form of every scaling law is set by its composition operator; it now states the answer under the stated axioms. The abstract called the fourth-cell test and the 80-domain expansion test registry-filed, and the caption of Figure 1 called the test of whether the two routes to the value two agree a filed test; each is a draft awaiting submission. Section 3 said that a replication with independent operator classification is planned; it has since been drafted, and it and the fourth-cell test are preregistered by timestamp. The epistemic-status line said that every law the programme names is under registered adversarial test; it now says so of Laws I to III and gives Laws IV and V their own status. The description of Figure 1 printed the sequential exponent of Paper II on its own; it now carries the whole of Paper II's row for that one model. The companion paper is cited without the version number it carried, 3.1. Reference [30] carried a wrong arXiv identifier and now carries arXiv:2206.01850; Section 11 said that Zhao (2022) derives the formula from network optimisation and now says producer-consumer dynamics, as Zhao's abstract does; references [16], [18] and [34] now carry their authors, titles and places of publication as their publishers record them; references [6] and [22] say that no official online source is available for the two books; and the He and Chen entry in Section 11 names the one record of theirs for 2003 at Crossref. Cited works whose official source could be confirmed are now linked to it. The withdrawal of Eden Engineering of 14 July 2026, which the note of 2 September 2026 below mentions, was reversed on 19 September 2026, and its version 6.3 is published. The abstract

said that the predictions themselves were timestamped in advance by dated public commit, and named the per-row classifications in the hash-pinned catalogue files of March 2026 beside the 12-domain extension manifest of 17 March 2026 and the book's two prediction appendices; it now says what Section 3 says: the predictions for this cohort are not timestamped in advance of their own fits, and the advance per-domain timestamp belongs to the 12-domain extension manifest and to the book's two prediction appendices. The abstract now also says that the fourth-cell test is preregistered by timestamp. The abstract, Section 5 and Section 8 said that the result is strong enough to publish as exploratory evidence; each now gives it the description the theory registration gives it, a structured prediction comparison under a publicly stated rule rather than a per-domain advance timestamp. The 19 of 25 now carries, at its first mention in each section, in the summary above the paper and in the table of Section 4, the [count withheld] of the rerun of 11 August 2026 that added the logarithmic candidate; until now that companion stood only in the note of 25 August 2026 below. Section 7.1 called the thirteen published exponents strong corroborating evidence; it now calls them corroborating evidence whose per-row sourcing is outstanding. Section 11 said that He and Zhang (2004) extend the derivation to heterogeneous branching networks and recover the same form under energy-minimisation assumptions; it now describes that work in the words of its abstract, and does the same for Maino et al. (2014), whose abstract names dynamic energy budget theory and states no exponent. The summary above the paper and Sections 12 and 13 no longer call the result or the sample exploratory; the limits they state stand. Section 3.2 links the four-family test to the programme's list of designs, where its words read as if the test were registered. The count of at least seven research groups in Section 11 stands, and now carries a clause that the He and Chen item could not be confirmed at a registry on 3 October 2026. Seven works cited in the text and absent from the reference list now have entries there, [36] to [42], each from its official record. In Section 12.1 a range of sections and the operator-to-family mapping are now written in words. No registered law, prediction, kill condition or dependency changes; no result is added; no date of any claim moves.

13 September 2026 (v3.11): The provenance note at the end has been corrected for accuracy. No claim, date, result or status has changed.

2 September 2026 (v3.10): a set of corrections that leaves every result standing: the Figure 1 caption and alt text carry the self-reference theorem in the subscripted form the notation spine uses, $\alpha = 1/(1 - \beta_L)$, noting that the artwork itself shows a bare β, and they identify the ARC Ceiling derivation set against it; reference [4] (Nemet 2009) now cites Research Policy; the 619-species estimate is credited to White and Seymour 2003, and its disagreement with row 26's compendium value is recorded next to the table; the summary description is no longer truncated; Eden Engineering is labelled withdrawn; and the first-published date reads 16 March 2026 as the register has it, 24 March having been the date of the draft.

25 August 2026 (v3.9): two dated clarifications that leave the science untouched. Wherever the 19/25 figure stood alone it now carries its corrected companion: the exploratory rerun of 11 August 2026, which added the logarithmic candidate, returns [count withheld], while the Cauchy-derived subset holds at 14/18 [in the original run; the rerun's count withheld], exactly as the abstract already states. The sentence describing pre-registration status is also marked as historical, the prospective fourth-cell test having since been prepared as a complete OSF draft that awaits human submission.

This paper and Laws IV and V, 3 October 2026. The theory has been stated with five laws since 24 September 2026. Law IV, the ARC Persistence Law, is carried by proposition P15, and Law V, the ARC Embedding Law, by propositions P12 and P21. The three propositions are registered; neither law is registered as a law, and neither has been tested. Their mathematics is set out in Laws IV and V: the

mathematics, version 1.0, of 3 October 2026. This paper classifies scaling families. It predicts the scaling family of each empirical domain from the domain's composition operator and compares the prediction with the family that AICc model selection returns. Its evidence tiers are reported separately, and the limits of its candidate set are stated in the paper. Its results do not test Law IV or Law V. No result in this paper is a deciding test of P15, P12 or P21. Naming the two laws changes nothing printed here: the results, their dates and their limits stand as they were.

Table 5. Where each block of the layout of version 3.15 stands in this edition.

Block in the layout of version 3.15	Where it stands in this edition
Series line above the title	Appendix F
Title, author lines and links	The title (the paper's established name and its question), the secondary line and the author block; the title of version 3.15, as recorded on OSF, in the first-page footnote; the website links are not set
Plate beneath the author line, and the subtitle	The first-page footnote (the paper's place in the programme) and Appendix F
Version line, with its links to the works the paper extends and to the validation code	The first-page footnote; Section 2 and Appendix F; the paragraph headed Code and data, after Section 15
Dated notes of versions 3.9 to 3.15	Appendix B.2, as printed
Abstract	The abstract, rewritten; its protocol summary opens Section 4, its account of the timing is Section 4.1, its numbers are in Section 6, its novelty sentence is in Section 1 and its keywords are in the PDF's metadata
Opening box on the paper's place in the programme	Appendix F
Note of 3 October 2026 on Laws IV and V	Section 12.3, restated; Appendix B.2, as printed
Summary above the paper, header and footer, and the structured and citation records, which the notes of versions 3.12 and 3.13 name	Elements of the website's page and of the citation files, not set in the paper
Section 1, The Question	Section 1
Section 2, The Theory; 2.1; 2.2 and its box 'The prediction'	Section 3; Section 3.1; Sections 3.2 and 3.3
Figure 1, its caption and its description	Figure 1, redrawn, at the end of Section 3; the original artwork's labels in Appendix B.4
Section 3, Prediction Protocol; its box 'Structured prediction comparison - not a blind test'; 3.1; 3.2	Section 4; Section 4.1; Section 4.2; Section 4.3
Section 4 and its table	Section 5 and Table 1
Section 5 and its table; 5.1	Section 6 and Table 2; Section 6.1
Section 6 and its table	Section 7 and Table 3
Section 7; 7.1 and its table; 7.2; 7.3; 7.4	Section 8; Section 8.1 and Table 4; Section 8.2; Section 8.3; Section 8.4
The box 'Note on the 3/4 exponent' in Section 7.1, which Section 11 called the Warning box	The note on the three-quarter exponent in Section 8.1

Block in the layout of version 3.15	Where it stands in this edition
Section 8	Section 9
Section 9; 9.1; 9.2; 9.3	Section 10; Section 10.1; Section 10.2; Section 10.3
Section 10; 10.1; 10.2	Section 11; Section 11.1; Section 11.2 and Appendix E
References [1] to [42]	The references, by author and year; Appendix E gives each one's number in version 3.15
Section 11, Related Work	Section 2
Section 12, Limitations; 12.1	Section 13; Section 13.1
Section 13, Implications	Section 14
Section 14, Conclusion	Section 15
Section 15, Reproducibility, and its list of companion papers	The paragraph headed Code and data, after Section 15; the companion papers in Appendix F
How to cite this paper	Not set: the first-page footnote gives the DOI
Epistemic-status line	Section 12.3, restated
Copyright and authorship line	The copyright and licence lines after the acknowledgements, in the programme's standard form
Standing covenant	The end of Section 13.1
Provenance note at the end	The acknowledgements, word for word

B.3 Earlier versions, as deposited on OSF

The paper's OSF component, 10.17605/OSF.IO/X6WA7, was created on 17 March 2026 at 01:16:28 UTC. It holds every deposit of the paper's PDF as a revision of one file, each dated by OSF's server (Table 6, read from OSF on 1 November 2026), and, from 15 August 2026, the paper's HTML, text and citation files beside it. A version's own date, printed in the file, is the author's date and can fall a day after the UTC time of its deposit. The first version, of 16 March 2026, was committed to the public repository that night (Appendix C); version 2.0 names version 1.1, of the same date, as the version it supersedes, and neither was deposited on the component. Versions 3.1 and 3.2 were not deposited as revisions of their own.

Table 6. The paper's PDF on OSF, revision by revision: the time of each deposit as OSF's server records it (UTC), and the version and date printed in the file.

Revision	Deposited on OSF (UTC)	Version printed in the file
1	17 March 2026, 01:17:44	2.0, 17 March 2026
2	17 March 2026, 01:26:25	2.0, 17 March 2026
3	20 March 2026, 17:12:42	2.0, 17 March 2026, first published 16 March 2026
4	24 March 2026, 16:01:31	3.0, 24 March 2026
5	4 July 2026, 15:42:44	3.0, 24 March 2026
6	13 August 2026, 21:49:51	3.3, 13 August 2026
7	14 August 2026, 02:28:48	3.4, 14 August 2026
8	15 August 2026, 02:04:42	3.4, 14 August 2026
9	15 August 2026, 23:21:47	3.5, 16 August 2026
10	16 August 2026, 00:27:08	3.6, 16 August 2026
11	17 August 2026, 00:58:17	3.6, 16 August 2026

Revision	Deposited on OSF (UTC)	Version printed in the file
12	17 August 2026, 01:52:53	3.6, 16 August 2026
13	17 August 2026, 08:42:07	3.6, 16 August 2026
14	20 August 2026, 23:54:11	3.6, 16 August 2026
15	22 August 2026, 09:13:16	3.7, 22 August 2026
16	22 August 2026, 12:08:22	3.7, 22 August 2026
17	23 August 2026, 06:35:47	3.7, 22 August 2026
18	23 August 2026, 07:47:43	3.8, 23 August 2026
19	1 September 2026, 08:41:58	3.9, 25 August 2026
20	1 September 2026, 14:53:58	3.9, 25 August 2026
21	1 September 2026, 18:34:44	3.9, 25 August 2026
22	2 September 2026, 20:42:03	3.10, 2 September 2026
23	13 September 2026, 10:10:00	3.11, 13 September 2026
24	20 September 2026, 08:46:56	3.11, 13 September 2026
25	4 October 2026, 09:00:05	3.12, 3 October 2026
26	4 October 2026, 14:02:02	3.13, 4 October 2026
27	4 October 2026, 17:26:29	3.14, 4 October 2026
28	5 October 2026, 13:10:53	3.15, 5 October 2026

The file of version 2.0 (revision 1) printed this note of its changes:

v2.0: expanded to 50 domains across five evidence tiers; replaced permissive R^2 criterion with proper AICc; honest miss analysis; tiered reporting

The file of version 3.8 (revision 18) printed a list headed ‘Corrections and version history’, limited to material scientific changes, whose notes read:

v3.6 (16 August 2026): the four-cell classification replaces the earlier three-family taxonomy throughout, including in the figure captions: a composition operator, a parameter family and a stability statement are distinct objects, and the earlier presentation collapsed them. The withdrawn below-0.2 per cent physics aggregate is removed from live use, and the companion references carry their own identifiers.

v3.4 (14 August 2026): the claim that the logarithmic functional equation governs diminishing returns is struck. Cauchy’s classification fixes which functional form a composition rule admits; it does not by itself establish which empirical regime any given system occupies. The classification result is preserved; the empirical inference drawn from it is not.

v3.1 (8 August 2026): the two displayed functional equations in Section 2.2 were corrected: the multiplicative and additive bullets displayed each other’s equations. The protocol, the Section 5 table and the validation code were unaffected.

v3.0 (24 March 2026): the space-filling overstatement was corrected by adding the conservation or optimisation constraint as a third necessary condition, and the debate over the mammalian three-quarter exponent is acknowledged rather than assumed away. The domain suite was expanded to fifty across five evidence tiers, the permissive fit criterion was replaced with AICc, and the misses are analysed rather than omitted.

The files of versions 3.0 to 3.6 also printed fuller working notes at their head, among them the notes of versions 3.2 (10 August 2026) and 3.3 (13 August 2026), which the list above does not carry. Those notes

name internal working records and website elements, which this paper does not print; each file is on OSF (Table 6). Version 3.3 added the section, Section 4.3 of this edition, that states the two families the six-model candidate set lacks.

B.4 The original artwork of Figure 1

Version 3.15 drew Figure 1 as a graphic in its web page. Its labels are recorded here as drawn, region by region, with each paper they name linked to its DOI; the name of the website data file in the source line is not printed. Appendix B.1 lists the changes made in redrawing the figure.

Title line. ARC Principle Architecture: $U = I \times g(R)$ with $\alpha = 1/(1-\beta)$

Top row. Base Intelligence; I (problem-solving at depth 0); $\times$; multiply; Recursive Amplification; $g(R) = [1 + (a/\alpha)R]^{\alpha}$; β: self-referential coupling (measured from data); Effective Capability; $U = I \times g(R)$ · measured as accuracy/error

Sequential card. Sequential Recursion; Each step builds on prior output; $\alpha_{\text{seq}} \approx 0.49$ $[-1.3, 2.9]$ (sub-linear); Error suppression: ~57% per depth

Parallel card. Parallel Recursion; Independent passes; no accumulation; $\alpha_{\text{par}} \approx 0$ (no scaling); Key result: form of recursion matters

ARC Bound card. ARC Bound; $\alpha \leq 2$: the proposed stable limit; A durably stable $\alpha > 2$ refutes the ARC Bound; $\alpha = 2.24$ RETRACTED (Paper IX §7)

Verification strip. Measured (v13): $\alpha_{\text{seq}} \approx 0.49 \cdot \alpha_{\text{par}} \approx 0 \cdot 6$ models · 4-layer blinding · $\beta \to \alpha$ identity: $R^2 = 1.000$ (analytical) · 30 Bernoulli ODE solutions

Falsification line. Falsification criteria recorded per claim · each sufficient to refute · Paper IX §4 and the register

Bottom strip. Cauchy Unification: $\alpha = d/(d+1)$ · 13/13 metabolic exponents match; six other domains miss, published · $\alpha = 1/(1-\beta)$ for recursion

Source line. Source: Papers I, II v13, III, VII, IX · OSF 10.17605/OSF.IO/6C5XB · [the name of a website data file]

C Priority record

Table 7 gives each dated record that bears on this paper's formulation and on the March 2026 comparison it reports, with its kind of timestamp; no kind is presented as stronger than it is. A commit date is written by whoever makes the commit and can be set by the author; a push time is recorded by GitHub's server when a commit reaches the public repository; an OSF time is recorded by OSF's server; a Bitcoin anchor fixes a document's hash in a block of the Bitcoin chain; a date written in a document is the author's own record. The public repository's history was later rewritten, so the commits of March 2026 carry new identifiers in its present history, under which they are named here, with the commit dates they carried when first pushed; the push times are GitHub's records of the pushes that first carried them. The versions of this paper are dated in Appendix B, and Appendix C.2 gives the two priority claims for this paper that version 3.15's text companion printed.

C.1 Dated records

Table 7. Dated records bearing on this paper, each with its kind of timestamp.

Date	Record	Identifier	Kind of timestamp
2 January 2026	*Infinite Architects*, first edition: the ARC Principle stated as a conjecture, $U = I \times R^2$; Appendix F's five numbered testable predictions; Appendix A, with further predictions in §A.3 and falsification criteria in §A.4	ISBN 978-1806056200; Eastwood (2026a)	Publication date of the first edition; a printed prior statement, not a preregistration
17 January 2026	Paper I first published, stating the ARC Principle formally as $U = I \times R^{\alpha}$	Eastwood (2026h); the programme's OSF project, 10.17605/OSF.IO/6C5XB	OSF's server dates: the project created at 08:17 UTC, and the paper uploaded to it at 09:18:20 UTC (version 1 of the file now in the project's archive folder); the paper's own date of first publication
13 February 2026	The Foundational paper first published, applying the form $\alpha = d/(d+1)$ to recursive capability; the form itself is earlier work (Section 2)	Eastwood (2026d)	Date written in the paper's own record
16 March 2026, 23:33:19 UTC	The first version of this paper, with the classification rule that assigns a scaling family to a domain, committed to the public repository; pushed at 23:36:28 UTC	Commit 23ce10a	Commit date, written by the committer; push time, recorded by GitHub's server
16 March 2026, 23:54:53 UTC	The fifty per-domain assignments committed together with their results, so that the 19 of 25 is a structured prediction comparison under a publicly stated rule, not a per-domain advance timestamp; pushed at 23:54:55 UTC	Commit 5435e77	Commit date; push time, recorded by GitHub's server
17 March 2026, 00:19:19 UTC	The 12-domain extension's twelve per-domain predictions and its written protocol, in a packet marked draft and not locked, committed 23 minutes before the run, which its run record dates 00:42:41 UTC; pushed at 00:19:20 UTC	Commit 8edaec0	Commit date; push time, recorded by GitHub's server; the run time, written in the run's own record

Date	Record	Identifier	Kind of timestamp
17 March 2026, 00:31:58 UTC	The temporal out-of-sample run's code and results first public together; pushed at 00:31:59 UTC	Commit b4f7e69	Commit date; push time, recorded by GitHub's server; the results file carries no time of its own, and nothing public shows that the pushed code is the code that ran
17 March 2026, 00:43:25, 00:47:30 and 00:56:10 UTC	The 12-domain run's result of 10 of 12; the conservative miss posture; the run relabelled a pilot dry run; pushed at 00:43:26, 00:47:32 and 00:56:12 UTC	Commits e5c22dd, f5e06a8 and c853277	Commit dates; push times, recorded by GitHub's server
17 March 2026, 01:16:28 and 01:17:44 UTC	This paper's OSF component created, and version 2.0 deposited on it (Appendix B.3)	10.17605/OSF.IO/X6WA7	OSF's server
17 March 2026, 02:01:59 to 02:07:11 UTC	The experiment folder deposited on this paper's OSF component: the suite's scripts, results and data from 02:01:59 UTC, then the extension's folder from 02:06:51 UTC, among it the twelve-row manifest of the 12-domain extension, as relabelled at 00:56 UTC with its twelve predictions unchanged (Eastwood 2026m), at 02:06:55 UTC, and its extracted data at 02:07:09 and 02:07:11 UTC	10.17605/OSF.IO/X6WA7	OSF's server; a dated deposit, not an OSF Registration
12 August 2026	The fourth-cell test and the replication with independent operator classification preregistered by timestamp, each first anchored at this block	Bitcoin block 962066	Bitcoin anchor
6 and 7 September 2026	The first ground of the relabelling of 17 March 2026, that a registry timestamp was required, corrected on 6 September, and the record re-read under the correction on 7 September; the second ground, that the run's own written rule to deposit its predictions with a registry before the domain data were extracted was not followed, stands	Theory registration, version 1.104 (Eastwood 2026m)	Dates written in the registration's text
8 September 2026	The theory registration, version 1.100, which registers propositions P18 and P9 in ARC-1, the register of Law I	10.17605/OSF.IO/P8CKQ	Registry record (OSF); updated on 13 September 2026 (1.102), 15 October 2026 (1.103) and 29 October 2026 (1.104)

C.2 The priority claims of version 3.15's text companion

Version 3.15's text companion printed four priority claims relevant to this paper. Two are carried here, with the account of the estimate of about 0.49 corrected; the other two bear on other papers of the programme and are left to them.

PC-036 (13 February 2026). The application of the bare exponent form $\alpha = d/(d+1)$ to recursive AI capability was first published in the Foundational paper on 13 February 2026 (Eastwood 2026d). The exponent form itself is prior work in the wider scaling-law literature, whose derivations Section 2 states each in its own terms; the priority is limited to the application to AI capability on that date.

PC-011 (2 January 2026). The ARC Principle equation, in its book form $U = I \times R^2$ and its paper form $U = I \times R^\alpha$, was first published as a conjecture in *Infinite Architects* on 2 January 2026 (Eastwood 2026a) and formally in Paper I on 17 January 2026 (Eastwood 2026h). The framing is conjecture throughout. The early estimate of α of about 2.24 is retracted. The best surviving estimate comes from the one of Paper II's five capability-arm models with clean, monotonic data, Gemini 3 Flash: about 0.49 by regression and 0.59 by endpoint estimate, whose bootstrap interval, from −1.3 to 2.9, makes it inconclusive (Eastwood 2026i). It is what survived the retraction of 2.24, not a correction of the same measurement. Paper II measures error suppression, so its estimates are error-decay estimates, not measurements of Law I's capability exponent: an inverse-error capability proxy gives the same exponent only when that metric and its power model are explicitly fixed. Whether $\alpha > 1$ remains an open prediction. The bare form $d/(d+1)$ is prior work.

D Notation and glossary

D.1 Symbols

Table 8. Symbols, with the place each is defined.

Symbol	Meaning	Defined in
f	a function satisfying one of the four equations	Theorem 1
x, y	arguments of f	Theorem 1
c	the real constant of a solution family	Theorem 1
a	the base of the exponential family, $a = e^c > 0$; in Figure 1, the rate constant of the amplification $g(R)$	Theorem 1, Figure 1
log	the natural logarithm	Theorem 1
U	a quantity that grows with recursive depth (the response); in Figure 1, effective capability	Section 3.2
R	recursive depth, the declared variable	Section 3.2
$s = \log R$	the logarithmic variable	Remark 2
K	a ceiling on output, the added assumption of the bounded class	Section 3.3
λ, R_0	the logistic rate and midpoint	Proposition 1
k, n	the Hill half-saturation scale and exponent ($n = 1$ for the Michaelis-Menten form)	Proposition 1
a, b in $a/(b+R)$	the two parameters of the hyperbolic decay, the seventh candidate fitted for domain 15	Section 4, step 4
F, ε	a positive response that obeys a composition rule only approximately, and a global bound on its logarithmic defect	Remark 3
d	effective transport dimension	Section 8.1
M	body mass	Section 8.1
L	a dimensionless linear size, against which total demand scales as L^d and the demand-weighted mean route length as L	Assumption 1

Symbol	Meaning	Defined in
n, k, R^2	sample size, parameter count and coefficient of determination in the v1 AIC approximation (not the Hill k and n)	Section 4.2, equation (1)
AICc	the corrected Akaike information criterion	Section 4.2
I, $g(R)$, α	base intelligence, the recursive amplification, and a scaling exponent: that of $g(R)$, and in the figure's bottom strip that of a physical system, $d/(d+1)$	Figure 1
β_L	the self-referential coupling, which sets $\alpha = 1/(1-\beta_L)$	Figure 1
γ, α_{crit}	the correction-leverage exponent of Law III, the ARC Ceiling, and its critical exponent $\alpha_{\text{crit}} = 1/(1-\gamma)$	Figure 1

D.2 Plain-language glossary

The paper's technical terms, in one place and in ordinary words.

Table 9. Plain-language glossary.

Term	In plain English
Composition operator	How consecutive applications of a system's recursive process combine. It decides which of Cauchy's equations, if any, the response obeys (Section 3.2).
Multiplicative, additive and bounded composition	Multiplicative: when depths multiply, outputs multiply. Additive: when depths add, outputs multiply. Bounded: output is held below a ceiling. The first two are exact rules (Definition 1); the third rests on an assumed ceiling and on dynamics of its own, not on Cauchy's equations, and the ceiling alone selects no form (Proposition 1).
Scaling family	The kind of curve a scaling law follows, such as a power law, an exponential or a saturation curve, before its parameters (the exponent, the base, the ceiling) are fitted.
Nearest comparator	For a published metabolic exponent, the value of $d/(d+1)$ for $d = 1$, 2 or 3 (that is, $1/2$, $2/3$ or $3/4$) that it lies closest to; a match means that the closest value is the one predicted. Lying closest to a value is not evidence that the exponent equals it (Section 8.1).
Evidence tier	One of the five groups in which the results of the 50-domain suite are reported, by strength of evidence, from the 25 empirical curve fits down to the 6 analytic identities. The tiers are reported separately and never combined into one number (Section 5).
Structured prediction comparison under a publicly stated rule	The theory registration's description of this paper's test: the rule that assigns a family to each domain was public before the per-domain predictions were committed together with their results, so predicting each family before its fit is true of the procedure and not of the timestamp. It is not a blind trial (Section 4.1).
Preregistered by timestamp	Said of a design whose text was fixed and dated before its test by a timestamp, such as a Bitcoin anchor or a dated private upload, and stated with its date and receipt. It does not mean registered: that word is kept for what the author submitted to OSF and OSF accepted.

Term	In plain English
Attestation class	The kind of record that dates an item: a public repository commit date, which the committer writes; a server's own timestamp, such as an OSF upload or a GitHub push; a Bitcoin anchor; or a date written inside a document. No kind is presented as stronger than it is.
Analytic identity	A domain whose data come from an exact formula, such as $E = mc^2$. Fitting it only checks that the fitter recovers the formula, and it carries no independent evidential weight (Section 8.3).
Marginal-conditioned permutation test	The primary significance test: the numbers of domains predicted and selected in each family are held fixed, and the pairing of predictions with selections is shuffled 400,000 times; the p value is the share of shuffles with at least as many matches as were observed (Section 4.2).

E Data sources for the 25 empirical domains

Table 10 gives the source of each of the 25 empirical domains of Table 2. Domain n was reference [n] of version 3.15, and that reference's data note is given here word for word, with each work it names set as a citation and its one range written with 'to'. Where version 3.15 recorded that a work has no official online source, or where none could be confirmed for the details it printed, the entry in the references says so.

Table 10. The data source of each of the 25 empirical domains of Table 2, with the data note of version 3.15.

#	Domain	Sources	Data note in version 3.15
1	Kleiber's Law (Metabolic Scaling)	Kleiber 1932	Reproduced in Dodds et al. (2001).
2	Urban Scaling (GDP vs Population)	Bettencourt et al. 2007	Data: BEA + Census 2006, 25 largest US MSAs.
3	Species-Area (Galapagos)	Johnson and Raven 1973	Data: R *faraway::gala* dataset, 30 islands.
4	Wright's Law (Solar PV)	Nemet 2009; Farmer and Lafond 2016	IRENA; Our World in Data.
5	Heaps' Law (Vocabulary)	Manning et al. 2008	Reuters RCV1 corpus.
6	Zipf's Law (Word Frequency)	Kučera and Francis 1967	Brown Corpus.
7	Learning Curve (Cigar Rolling)	Crossman 1959	Reanalysis: Newell and Rosenbloom (1981).
8	Moore's Law (Transistor Count)	Wikipedia n.d.	Public vendor specifications (Intel, AMD, TSMC).
9	Radioactive Decay (P-32)	NIST Nuclear Data Center n.d.	P-32 half-life: 14.29 days. Exact theoretical values.
10	Gutenberg-Richter (Earthquakes)	U.S. Geological Survey n.d.	IRIS compilation. Global seismicity catalogue.
11	Bacterial Growth (*E. coli*)	Sezonov et al. 2007	None
12	O_2-Haemoglobin Curve	Severinghaus 1979	None
13	Epidemic SIR (Ebola 2014)	World Health Organization 2015; WHO Ebola Response Team 2014	WHO/CDC Ebola Situation Reports (2014 to 2015); NEJM Ebola Response Team.
14	Amdahl's Law (CPU Scaling)	Amdahl 1967	See also Hill and Marty (2008).
15	Muscle Force-Velocity	Hill 1938	None
16	Facebook MAU Growth	Zhang et al. 2015	Meta SEC filings
17	Brownian Diffusion (MSD)	Catipovic et al. 2013	None
18	Horton's Law (Stream Numbers)	Singh et al. 2021	None
19	Neural Scaling Laws (LLM Loss)	Kaplan et al. 2020	None

#	Domain	Sources	Data note in version 3.15
20	Time Crystal Order (Rydberg Gas)	Shen et al. 2025	See also Kongkhambut et al. (2024).
21	Stellar Mass-Luminosity	Torres et al. 2010; Eker et al. 2018	IAU 2015 nominal solar values (Prša et al. 2016).
22	Heart Rate vs Body Mass	Stahl 1967; Calder 1984; Schmidt-Nielsen 1984	None
23	Rent's Rule (VLSI Pin Count)	Landman and Russo 1971; Christie and Stroobandt 2000; Stroobandt 2001	None
24	Taylor's Power Law	Taylor 1961; Taylor and Woiwod 1980	None
25	Hack's Law (Stream Length)	Hack 1957; Mueller 1973; Rigon et al. 1996	None

Version 3.15 numbered its references, and the notes of Appendix B.2 name some of them by number. References [1] to [25] are the domains of Table 10, in the same order. References [26] to [35] were the compendium literature on metabolic scaling behind Table 4: [26] West et al. 1997; [27] Banavar et al. 1999; [28] Banavar et al. 2010; [29] Demetrius 2003; [29b] Demetrius 2006; [29c] Demetrius and Tuszynski 2010; [30] Zhao 2022; [31] Bettencourt 2013; [32] White and Seymour 2003; [33] Glazier 2005; [34] Glazier 2022; [35] Kolokotrones et al. 2010. References [36] to [42] were works its text cites: [36] Cauchy 1821; [37] Bingham and Ostaszewski 2015; [38] He and Chen 2003; [39] He and Zhang 2004; [40] Maino et al. 2014; [41] West 2017; [42] Aczél 1966.

F The paper's place in the programme

Placement. Version 3.15 printed this statement of the paper's place in the programme beneath its author line: "The classification behind Law I's form: under the stated axioms, Cauchy's functional equations force a power law for multiplicative composition and an exponential for additive composition, with saturation compatible with bounded composition and exhaustiveness open." Within Recursive Dynamics, the field the author founded, and its founding theory, the ARC Theory, the paper informs Law I, the ARC Principle, a law under test: it tests the classification behind the law's form, not the law's exponent. It decides no proposition of Laws II to V, whose status Section 12.3 gives; it carries no proposition of the Eden Protocol, and it makes no claim about the HARI Treaty, the Draft Treaty on Hardware-Aligned Recursive Intelligence.

The propositions it informs. Version 1.104 of the theory registration places five propositions in ARC-1, the register that tests the first law: P1, P2, P8, P9 and P18. This paper informs two of them. P18, the composition operator predicts the family, reads: "Classified in advance by how a domain composes its inputs and outputs, the scaling family that best fits that domain's data is the family the classification predicts, and not one chosen after the curve is seen." P9, the cross-domain form, reads: "Where an effective dimension can be assigned to a domain, the growth exponent follows the dimensional relation with the dimension supplied by that domain's equation of state." The primary comparison of Section 6 bears on P18, and the published exponents of Section 8.1 on P9; neither is a registered verdict (Section 12.2). The programme's Operational Definitions, version 2.0.0, state Cauchy's multiplicative equation as EQ-16, which informs P18, and record that what has to be checked in each application is whether its regularity conditions hold over the domain claimed.

Frame key. This paper carries a result: under a stated regularity condition and in a declared variable, Cauchy's functional equations force a power law under multiplicative composition and an exponential

under additive composition (Theorem 1), turning a two-hundred-year-old identity into a physically testable prediction about the shapes empirical scaling curves may take. For bounded composition no form follows from the equations: a saturation curve rests on a ceiling on output, which none of the four equations supplies, and on dynamics of its own (Proposition 1). Inside the ARC Theory it gives the classification behind the form of Law I, the ARC Principle, and is one of the ARC/Eden experiments, the mathematical spine that ties composition constraints to observed functional forms. Section 2 of *The Eden Vision* (Section II.A.8 in its version 2.9 and earlier) enumerates the nearest prior documents to the programme's doctrine of values embedded at the foundation, with the differential against each; this paper's own antecedents are set out in Section 2.

Series line and tagline. The paper's series line reads: The ARC Theory (the Theory of Artificial Recursive Creation) · ARC/Eden experiments · Paper VII. Its tagline reads: *ARC/Cauchy scaling classification: a structured prediction comparison across 50 domains in five evidence tiers, from the ARC/Eden experiments of the Theory of Artificial Recursive Creation*.

Extends. This paper extends the framework paper, *On the Origin of Scaling Laws*, and the Foundational Paper.

Companion papers. The programme's companion papers, each linked to its OSF DOI: Paper I; the Foundational Paper; Paper II; Paper III; On the Origin of Scaling Laws; Papers IV.a, IV.b, IV.c and IV.d; Paper V; Paper VI; Paper VIII; Paper IX; Eden Engineering; Eden Vision; the Executive Summary; and the Master Table of Contents.